

# Quantum confinement

## 6

In this chapter we give an overview of the optical properties of quantum-confined semiconductors. These are artificial structures in which the electrons and holes are confined in one or more directions. The structures that we consider generally have sizes in the nanometre range, and may thus be considered as examples of **nanostuctures**. We concentrate mainly on quantum wells, in which the confinement is just in one dimension, since these illustrate the main points most clearly. We then give an introduction to the physics of quantum dots, which is a subject that has advanced very rapidly since the first edition of this book. As we shall see, quantum wells and dots have very interesting optical properties that readily lend themselves to applications in opto-electronics. Furthermore, the physical principles that we shall study here for inorganic semiconductors can readily be adapted to other types of quantum-confined systems, such as the carbon nanostructures considered in Section 8.5.

The optical properties of quantum-confined semiconductors are derived from the physics of interband absorption, excitons, and interband luminescence discussed in Chapters 3–5. We presuppose here that these subjects have been fully assimilated, and the new material in this chapter provides a good opportunity to practise and develop the principles that have been learnt previously.

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## 6.1 Quantum-confined structures

The optical properties of solids do not usually depend on the size of the crystal. Rubies, for example, have the same red colour irrespective of how big they are. This statement is only true as long as the dimensions of the crystal are large. If we make very small crystals, then the optical properties do in fact depend on the size. A striking example of this is semiconductor-doped glasses. As discussed in Section 6.8.2, these contain very small semiconductor nanocrystals within a colourless glass host, and the colour can be altered just by changing the size of the crystals.

The size dependence of the optical properties in very small crystals is a consequence of the **quantum confinement** effect. The Heisenberg uncertainty principle tells us that if we confine a particle to a region of the  $x$  axis of length  $\Delta x$ , then we introduce an uncertainty in its momentum given by:

$$\Delta p_x \sim \frac{\hbar}{\Delta x}. \quad (6.1)$$

If the particle is otherwise free, and has a mass  $m$ , the confinement in

**Table 6.1** Number of degrees of freedom tabulated against the dimensionality of the quantum confinement. The final column shows the functional form of the density of states for free electrons.

| Structure                 | Quantum confinement | Number of free dimensions | Electron density of states |
|---------------------------|---------------------|---------------------------|----------------------------|
| Bulk                      | none                | 3                         | $E^{1/2}$                  |
| Quantum well/superlattice | 1-D                 | 2                         | $E^0$                      |
| Quantum wire              | 2-D                 | 1                         | $E^{-1/2}$                 |
| Quantum dot/box           | 3-D                 | 0                         | discrete                   |

the  $x$  direction gives it an additional kinetic energy of magnitude

$$E_{\text{confinement}} = \frac{(\Delta p_x)^2}{2m} \sim \frac{\hbar^2}{2m(\Delta x)^2}. \quad (6.2)$$

This confinement energy will be significant if it is comparable to or greater than the kinetic energy of the particle due to its thermal motion in the  $x$  direction. This condition may be written:

$$E_{\text{confinement}} \sim \frac{\hbar^2}{2m(\Delta x)^2} > \frac{1}{2}k_B T, \quad (6.3)$$

and tells us that quantum size effects will be important if

$$\Delta x \lesssim \sqrt{\frac{\hbar^2}{mk_B T}}. \quad (6.4)$$

This is equivalent to saying that  $\Delta x$  must be comparable to or smaller than the de Broglie wavelength  $\lambda_{\text{dB}} \equiv h/p_x$  for the thermal motion.

The criterion given in eqn 6.4 gives us an idea of how small the structure must be if we are to observe quantum confinement effects. At room temperature, we find that we must have  $\Delta x \lesssim 5 \text{ nm}$  for an electron in a typical semiconductor with  $m_e^* = 0.1m_0$ . Thus a 'thin' semiconductor layer of thickness  $1 \mu\text{m}$  is not thin by the standards of the electrons. It is in fact a bulk crystal that would not exhibit any quantum size effects except at extremely low temperatures. (See Exercise 6.1.) To observe quantum size effects we need thinner layers.

Quantum-confined structures are generally classified by their dimensionality. Table 6.1 summarizes the three basic types of quantum-confined structures that can be produced, for which the following nomenclature is usually adopted:

- **quantum wells:** 1-D confinement;
- **quantum wires:** 2-D confinement;
- **quantum dots:** 3-D confinement.

Table 6.1 also lists the number of degrees of freedom associated with the type of quantum confinement. The electrons and holes in bulk semiconductors are free to move within their respective bands in all three directions, giving them three degrees of freedom, and hence three-dimensional

The principle of equipartition of energy tells us that we have a thermal energy of  $k_B T/2$  for each degree of freedom of the motion.

Certain types of quantum well structures are called superlattices (see Section 6.2), while quantum dots are sometimes called quantum boxes.

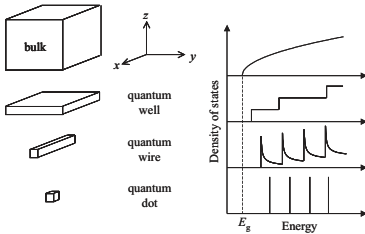


Fig. 6.1 Schematic representation of quantum wells, wires, and dots. The generic shape of the density of states function for electrons in the conduction band of a semiconductor with band gap  $E_g$  is shown for each type of structure.

(3-D) physics. The electrons and holes in a quantum well, by contrast, are confined in one direction, and therefore only have two degrees of freedom. This means that they effectively behave as 2-D materials. Similarly, quantum wire structures have 1-D physics, while quantum dots have ‘0-D’ physics. This last point means that the motion of the electrons and holes is quantized in all three dimensions, so that they are completely localized in the quantum dot. Quantum wells, wires, and dots are thus all examples of **low-dimensional structures**.

The quantization of the motion of the electrons and holes has two main consequences:

- (1) The energy of a particle at rest is increased by the quantum confinement energy.
- (2) The functional form of the density of states is changed.

These two points will be discussed at length throughout the chapter, and at this stage it is useful just to make some general comparisons, as illustrated schematically in Fig. 6.1.

The conduction band electrons in a bulk semiconductor can have any energy above the band gap energy  $E_g$  and the density of states is proportional to  $(E - E_g)^{1/2}$ . This is a consequence of the free motion in all three dimensions. The density of states for a quantum well is determined by the 2-D free motion and the shift of the energy due to the quantum confinement. As shown in Exercise 6.3, the density of states is independent of the energy, and so we have a sequence of steps in the density of states for each quantized level. Note that the band edge is effectively shifted to higher energy by the quantized energy for the quantum-confined motion in the third direction.

The argument can be repeated for 1-D quantum wire and 0-D quantum dot systems. In the case of quantum wires, the density of states has an  $E^{-1/2}$  dependence (see Exercise 6.4) which leads to peaks at each

See, for example, eqn 3.16, which shows that the density of states for free electrons varies as  $E^{1/2}$  in three dimensions. For an electron in the conduction band of a semiconductor, the energy must be measured relative to the bottom of the conduction band. Note that the  $(E - E_g)^{1/2}$  dependence only applies within the parabolic band approximation.

new quantized state as shown in Fig. 6.1. In quantum dots the motion is quantized in all three directions and there are no continuous bands at all. The density of states consists of a series of Dirac  $\delta$ -functions at each quantized level, as illustrated in Fig. 6.1. In this sense, quantum dots behave like ‘artificial atoms’ in which the electrons have discrete energies rather than continuous bands as is the norm in solid-state physics.

The very small crystal dimensions required to observe quantum confinement effects have to be produced by special techniques.

- Quantum wells are made by techniques of advanced epitaxial crystal growth. This will be explained in Section 6.2.
- Quantum wires are made by lithographic patterning of quantum well structures, or by epitaxial growth on patterned substrates.
- Quantum dots can be made by lithographic patterning of quantum wells or by spontaneous growth techniques, as discussed in Section 6.8.

In the sections that follow, we mainly concentrate on quantum well structures. This is because they illustrate the physical principles very well, and are already widely used in many commercial opto-electronic devices. We also briefly consider the optical properties of quantum dots. We do not mention quantum wires further here, due to the difficulties associated with making them, although we shall return to 1-D materials when we consider carbon nanotubes in Section 8.5.3.

## 6.2 Growth and structure of quantum wells

Semiconductor quantum wells are examples of **heterostructure** crystals. Heterostructures are artificial crystals that contains layers of different materials grown on top of a thicker substrate crystal. The structures are made by the specialized epitaxial crystal growth techniques introduced previously in Section 5.4.1. The two most important ones are **molecular beam epitaxy** (MBE) and **metal–organic chemical vapour deposition** (MOCVD), which is also called **metal–organic vapour-phase epitaxy** (MOVPE) by some authors. The layer thicknesses of the crystals grown by these techniques can be controlled with atomic precision. This makes it easy to achieve the thin layer thicknesses required to observe quantum confinement of the electrons in a semiconductor at room temperature.

Figure 6.2(a) shows a schematic diagram of the simplest type of quantum well that can be grown. In this particular case, a GaAs/AlGaAs structure grown on a GaAs substrate is shown. The structure consists of a GaAs layer of thickness  $d$  sandwiched between much thicker layers of the alloy semiconductor AlGaAs.  $d$  is chosen so that the motion of the electrons in the GaAs layer is quantized according to the criterion given in eqn 6.4. We set up axes so that the  $z$  axis corresponds to the crystal

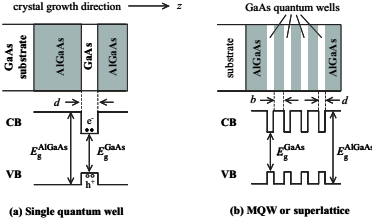


Fig. 6.2 (a) Schematic diagram of a single GaAs/AlGaAs quantum well. The quantum well is formed in the thin GaAs layer sandwiched between AlGaAs layers which have a larger band gap. The lower half of the figure shows the spatial variation of the conduction band (CB) and the valence band (VB). (b) Schematic diagram of a GaAs/AlGaAs multiple quantum well (MQW) or superlattice structure. The distinction between an MQW and a superlattice depends on the thickness  $b$  of the barrier separating the quantum wells.

growth direction, while the  $x$  and  $y$  axes lie in the plane of the layers. We thus have quantized motion in the  $z$  direction, and free motion in the  $x, y$  plane.

The bottom half of Fig. 6.2(a) shows the spatial variation of the conduction and valence bands that corresponds to the change of the composition along the  $z$  direction. The band gap of AlGaAs is larger than that of GaAs, and the bands line up so that the lowest conduction and valence band states of the GaAs lie within the gap of the AlGaAs. This means that electrons in the GaAs layer are trapped by potential barriers at each side due to the discontinuity in the conduction band. Similarly, holes are trapped by the discontinuity in the valence band. These barriers quantize the states in the  $z$  direction, but the motion in the  $x, y$  plane is still free. We thus effectively have a 2-D system in which the electrons and holes are quantized in one direction and free in the other two.

Epitaxial techniques are very versatile, and they allow the growth of a great variety of quantum well structures. Figure 6.2(b) shows one such variant derived from the single well structure shown in Fig. 6.2(a). The crystal consists of a series of repeated GaAs quantum wells of width  $d$  separated from each other by AlGaAs layers of thickness  $b$ . This type of structure is either called a **multiple quantum well (MQW)** or a **superlattice**, depending on the parameters of the system. The distinction depends mainly on the value of  $b$ .

MQWs have large  $b$  values, so that the individual quantum wells are isolated from each other, and the properties of the system are essentially the same as those of single quantum wells. They are often used in optical applications to give a usable optical density. It would be very difficult to measure the optical absorption of a single 10 nm thick quantum well, simply because there is so little material to absorb the light. By growing many identical quantum wells, the absorption will increase to a measurable value.

Superlattices, by contrast, have much thinner barriers. The quantum wells are then coupled together by tunnelling through the barrier, and new extended states are formed in the  $z$  direction. Superlattices have additional properties over and above those of the individual quantum wells.

Quantum well structures of the type shown in Fig. 6.2 can only be made if the properties of the constituent compounds are favourable to the formation of the artificial crystals. We have already noted in Section 5.4.1 that the unit cell size of GaAs and AlAs (and hence also the  $\text{Al}_x\text{Ga}_{1-x}\text{As}$  alloy) are almost identical: see Fig. 5.11. This means that both the GaAs and AlGaAs layers in the quantum well structure are lattice matched to the GaAs substrate, enabling dislocation-free crystals to be grown.

In recent years it has been realized that it is also possible to make quantum wells from materials with different unit cell sizes. This allows much more flexibility in the combinations of materials that can be used. The mismatch in the lattice constants introduces strain into the structure, but high-quality crystals can still be grown provided the total layer thickness is kept below a critical value. We briefly mention the application of these non-lattice-matched quantum wells in light-emitting diodes and laser diodes in Section 6.6.

## 6.3 Electronic levels

The wave functions and energies of the quantized states in the conduction and valence bands of a quantum well can be calculated by using Schrödinger's equation and the effective mass approximation. Fortunately, we do not have to solve the Schrödinger equation in three dimensions because the problem separates naturally between the free motion in the  $x, y$  plane and the quantized motion in the  $z$  direction. In this section, we first explain how this separation of variables works, and then go on to discuss the quantized states in the  $z$  direction in two different approximations. We treat the electron and holes separately here, postponing till Section 6.4 the discussion of the effects of electron-hole Coulomb interaction that leads to the formation of excitons.

### 6.3.1 Separation of the variables

The electrons and holes in a quantum well layer are free to move in the  $x, y$  plane but are confined in the  $z$  direction. This allows us to write the wave functions in the form:

$$\Psi(x, y, z) = \psi(x, y) \varphi(z), \quad (6.5)$$

and then solve separately for  $\psi(x, y)$  and  $\varphi(z)$ . The states of the system are described by two parameters: a wave vector  $\mathbf{k}$  to specify the free motion in the  $x, y$  plane, and a quantum number  $n$  to indicate the energy level for the  $z$  direction. The total energy is then obtained by adding

together the separate energies for the  $z$  and  $x, y$  motion, according to:

$$E^{\text{total}}(n, \mathbf{k}) = E_n + E(\mathbf{k}), \quad (6.6)$$

where  $E_n$  is the quantized energy of the  $n$ th level.

We can deal with the  $x, y$  plane motion very quickly. Since the motion is free, the electron and hole wave functions are described by plane waves of the form:

$$\psi_{\mathbf{k}}(x, y) = \frac{1}{\sqrt{A}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (6.7)$$

where  $\mathbf{k}$  is the wave vector of the particle, and  $A$  is the normalization area. Note that  $\mathbf{k}$  and  $\mathbf{r}$  only span the two-dimensional  $x, y$  plane here. The energy corresponding to this motion is just the kinetic energy determined by the effective mass:

$$E(\mathbf{k}) = \frac{\hbar^2 \mathbf{k}^2}{2m^*}. \quad (6.8)$$

The total energy for an electron or hole in the  $n$ th quantum level is therefore given by:

$$E^{\text{total}}(n, \mathbf{k}) = E_n + \frac{\hbar^2 \mathbf{k}^2}{2m^*}. \quad (6.9)$$

### 6.3.2 Infinite potential wells

The calculation of the wave functions and energies for the quantized states in the  $z$  direction is determined by the spatial dependence of the conduction and valence bands. We begin by considering the simplest case in which we assume that the confining barriers are infinitely high. This allows us to model the states by those of a 1-D potential well with infinite barriers, as shown in Fig. 6.3.

We consider a quantum well of thickness  $d$  and define position and energy coordinates such that the potential is zero for  $-d/2 < z < +d/2$  (i.e. inside the well) and  $\infty$  elsewhere, as indicated in Fig. 6.3. The choice of  $z = 0$  at the centre of the well is convenient, since it corresponds to the symmetry axis of the potential. The Schrödinger equation within the well is:

$$-\frac{\hbar^2}{2m^*} \frac{d^2 \varphi(z)}{dz^2} = E \varphi(z). \quad (6.10)$$

Since the barriers are infinitely high, there is no probability that the particle can tunnel out of the well. The solutions of eqn 6.10 are therefore subject to the boundary condition that  $\varphi = 0$  at the interfaces.

It can be checked by substitution that the normalized wave functions that satisfy eqn 6.10 and the boundary conditions are of the form:

$$\varphi_n(z) = \sqrt{\frac{2}{d}} \sin \left( k_n z + \frac{n\pi}{2} \right), \quad (6.11)$$

where  $n$  is an integer that gives the quantum number of the state, and

$$k_n = \frac{n\pi}{d}. \quad (6.12)$$

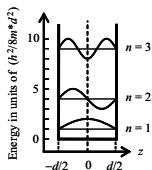


Fig. 6.3 The infinite one-dimensional potential well. The first three energy levels and corresponding wave functions are shown.

In a bulk semiconductor like GaAs, the heavy and light hole states are degenerate at  $k = 0$ . This is a consequence of the high symmetry of the cubic lattice. The lifting of this degeneracy is caused by the difference in the effective masses, but it can also be seen as a consequence of the lower symmetry of the quantum well. The bulk crystals are isotropic, but quantum wells are not: the  $z$  direction is physically distinguishable from the other two. As explained in Section 1.5.1, we therefore expect certain degeneracies to be lifted, in the same way that a magnetic field splits the heavy- and light-hole states of bulk GaAs via their different magnetic energies.

This form of wave function describes a standing wave inside the well with nodes at the interfaces. The energy that corresponds to the  $n$ th level is given by

$$E_n = \frac{\hbar^2 k_n^2}{2m^*} = \frac{\hbar^2}{2m^*} \left( \frac{n\pi}{d} \right)^2. \quad (6.13)$$

The wave functions of the first three levels are shown in Fig. 6.3. Equation 6.13 describes an infinite ladder of levels with quantization energy increasing in proportion to  $n^2$  in units of  $(\hbar^2 \pi^2 / 2m^* d^2)$ . The ground state is the  $n = 1$  level, and the levels of higher  $n$  are the excited states of the system.

The energies of the first two levels for an electron with  $m^* = 0.1m_0$  in a 10 nm quantum well are 38 meV and 150 meV respectively. These values should be compared to the thermal energy  $k_B T$ , which is 25 meV at room temperature. It is clear that the quantization energy is greater than the thermal energy at room temperature, and thus that the quantum description of the motion is appropriate. The comparison of the quantization energies to the thermal energy gives a criterion to decide whether a particular quantum well will in fact exhibit quantum effects at a particular temperature. This criterion can be compared to the one based on the Heisenberg uncertainty principle given in eqn 6.4. It is easy to show that the two criteria predict a crossover from classical to quantum behaviour at roughly the same value of  $d$ . (See Exercise 6.2.)

Although real semiconductor quantum wells have finite barriers, the infinite barrier model is a good starting point for a discussion of their properties. The accuracy of the model will be best for states with small quantization energies in material combinations that give rise to high barriers at the interfaces. A few useful general points emerge from the analysis:

- (1) The energy of the levels is inversely proportional to the effective mass and the square of the well width. This means that low mass particles in narrow quantum wells have the highest energies.
- (2) Since the energy depends on the effective mass, the electrons, heavy holes, and light holes will all have different quantization energies. In the valence band, the heavy holes will have the lowest energy, and are dominant in most situations because they form the ground-state level.
- (3) The wave functions can be identified by their number of nodes, i.e. the number of zero crossings within the well. It is evident from Fig. 6.3 that the  $n$ th level has  $(n - 1)$  nodes.
- (4) The states are also labelled by their **parity** with respect to inversion about the centre of the well, that is, whether  $\varphi(-z) = +\varphi(z)$  (even parity) or  $\varphi(-z) = -\varphi(z)$  (odd parity). States of odd  $n$  have even parity, and vice versa.

These points also apply to more realistic models of quantum wells in which the barriers at the interfaces are only of finite height. As we shall see below, the infinite well model overestimates the quantization energy.

In real quantum wells with finite barriers, the particles are able to tunnel into the barriers to some extent, and this allows the wave functions to spread out further, thereby reducing the confinement energy.

### 6.3.3 Finite potential wells

Figure 6.4 shows the band diagram of a more realistic quantum well which has a finite potential barrier of height  $V_0$  at each interface. There are now only a finite number of bound states with energy  $E < V_0$ . These bound states are labelled by a quantum number  $n$ , and it can be shown that there is always at least one, no matter how small  $V_0$  is: see Exercise 6.5.

The Schrödinger equation within the quantum well is the same as before (eqn 6.10). We therefore have sine and cosine solutions of the form:

$$\varphi_w(z) = C \sin(kz) \quad (6.14)$$

and

$$\varphi_w(z) = C \cos(kz), \quad (6.15)$$

where

$$\frac{\hbar^2 k^2}{2m_w^*} = E. \quad (6.16)$$

Note that we have added a subscript 'w' to the effective mass to clarify that it is the value for the semiconductor used in the quantum well layer. By comparison with eqn 6.11 and also Fig. 6.3, we see that the bound states with odd values of  $n$  have cosine solutions, while those with even  $n$  have sine solutions with a node at  $z = 0$ . The wave functions given in eqns 6.14 and 6.15 are valid for  $-d/2 \leq z \leq +d/2$ .

We now consider the extension of the wave functions into the barrier regions. This occurs because the finite potential discontinuity allows the electrons and holes to tunnel into the barriers. We therefore no longer have nodes at the interfaces. The Schrödinger equation in the barrier regions is given by:

$$-\frac{\hbar^2}{2m_b^*} \frac{d^2 \varphi(z)}{dz^2} + V_0 \varphi(z) = E \varphi(z), \quad (6.17)$$

where  $m_b^*$  is the effective mass of the barrier material. In general,  $m_b^*$  and  $m_w^*$  will not be the same because the materials that constitute the quantum well and barrier regions have different band structures. The solutions of eqn 6.17 are exponentials of the form:

$$\varphi_b(z) = C' e^{\pm \kappa z} \quad (6.18)$$

where  $\kappa$  satisfies:

$$\frac{\hbar^2 \kappa^2}{2m_b^*} = V_0 - E. \quad (6.19)$$

For bound states we require that the solutions decay in the barrier, and so we choose  $\varphi(z) = C' \exp(-\kappa z)$  for  $z \geq +d/2$  and  $\varphi(z) = C' \exp(+\kappa z)$  for  $z \leq -d/2$ .

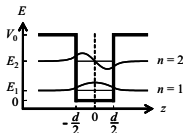


Fig. 6.4 First two bound states of a finite potential well of depth  $V_0$  and width  $d$ .

The wave functions and energies of the bound states can be found by applying the appropriate boundary conditions at the interfaces. These tell us that both the wave function  $\varphi(z)$  and the particle flux  $(1/m^*)d\varphi/dz$  must be continuous at  $\pm d/2$ . Hence we must have

$$\varphi_w(\pm d/2) = \varphi_b(\pm d/2), \quad (6.20)$$

and

$$\frac{1}{m_w^*} \left( \frac{d\varphi_w}{dz} \right)_{z=\pm d/2} = \frac{1}{m_b^*} \left( \frac{d\varphi_b}{dz} \right)_{z=\pm d/2}. \quad (6.21)$$

The wave functions must have symmetry about  $z = 0$ , so we just concentrate on  $z = +d/2$ . We consider the solutions with cosine solutions in the quantum wells first. The wave function continuity requires that

$$C \cos(kd/2) = C' \exp(-\kappa d/2), \quad (6.22)$$

while flux continuity requires that

$$-C \frac{k}{m_w^*} \sin(kd/2) = -C' \frac{\kappa}{m_b^*} \exp(-\kappa d/2). \quad (6.23)$$

On dividing eqn 6.23 by eqn 6.22, we find:

$$\tan(kd/2) = \frac{m_w^* \kappa}{m_b^* k}. \quad (6.24)$$

On following a similar procedure for the solutions with sine solutions in the quantum well, we find:

$$\tan(kd/2) = -\frac{m_b^* k}{m_w^* \kappa}. \quad (6.25)$$

On substituting the values of  $k$  and  $\kappa$  from eqns 6.16 and 6.19, we can now solve for the energy  $E$  of the bound states, and hence find the wave functions. Unfortunately, there is no analytic solution for  $E$ . The equations have to be solved numerically or graphically. An example of how this is done is given in Example 6.1.

It is useful to make a few general observations about the solutions, as we did for the case of the infinite well in the previous section.

- (1) The spreading of the wave functions into the barrier by tunnelling increases  $k$  and hence reduces the quantum confinement energy compared to a well with infinite barriers.
- (2) The decay constant can be found by substituting  $E_n$  for  $E$  in eqn 6.19. This means that the levels near the top of the well with  $E_n$  close to  $V_0$  tunnel more into the barrier regions because they have a smaller decay constant.
- (3) The eigenstates can be identified by the number of nodes, just as for infinite wells. The  $n$ th bound state has  $(n-1)$  nodes. The potential energy has even symmetry about  $z = 0$ , and so the eigenstates have well-defined parities.

The possibility of tunnelling into the barriers gives rise to a whole series of electronic and opto-electronic quantum well tunnelling devices.

Figure 6.4 sketches the wave functions of a typical finite well with two bound states. The similarity between these wave functions and the first two states of the infinite well shown in Fig. 6.3 is apparent. The main difference is that the wave functions of the finite well spread out more by tunnelling into the barrier, whereas the wave functions of the infinite well stop abruptly at the interfaces.

It is useful to compare directly the predictions of the finite and infinite well models. Table 6.2 tabulates the energies of the bound states of a 10 nm GaAs quantum well with  $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$  barriers for the two models. In all cases, the infinite well model overestimates the quantization energy. The discrepancy gets worse for the higher levels. Note that the quantization energies of the heavy holes are smaller than those of the electrons because of their heavier effective mass. Note also that the separation of the first two electron levels is more than three times the thermal energy at room temperature, where  $k_{\text{B}}T \sim 25$  meV. This confirms that we expect to observe 2-D physics for the electrons at 300 K. The quantum confinement of the heavy holes is less good, but is still acceptable since  $E_2 - E_1$  is comparable to  $k_{\text{B}}T$  at 300 K. Although the infinite well model overestimates the confinement energies, it is a useful starting point for the discussion of the physics because of its simplicity.

**Table 6.2** Bound states of a 10 nm GaAs/ $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$  quantum well calculated using the finite and infinite well models. The states are labelled by the particle type (e for electron, hh for heavy hole and lh for light hole) and by the quantum number  $n$ . All energies are in meV.

| State | Finite well | Infinite well |
|-------|-------------|---------------|
| e1    | 32          | 57            |
| e2    | 120         | 227           |
| e3    | 247         | 510           |
| hh1   | 7           | 11            |
| hh2   | 30          | 44            |
| hh3   | 66          | 100           |
| hh4   | 112         | 177           |
| lh1   | 21          | 40            |
| lh2   | 78          | 160           |

### Example 6.1

Calculate the energy of the first electron bound state in a GaAs/AlGaAs quantum well with  $d = 10$  nm and  $V_0 = 0.3$  eV. Take  $m_w^* = 0.067m_0$  and  $m_b^* = 0.092m_0$ . Compare this value to the one calculated for an infinite quantum well.

#### Solution

The first bound state has a maximum at  $z = 0$ , and so we look for the solutions with cosine wave function in the well region. By making the substitution  $x = kd/2$ , we can use eqns 6.16 and 6.19 to recast eqn 6.24 in the form:

$$x \tan x = \left( \frac{m_w^*}{m_b^*} \right)^{1/2} \sqrt{\xi - x^2}, \quad (6.26)$$

where

$$\xi = \frac{m_w^* d^2 V_0}{2\hbar^2}, \quad (6.27)$$

and

$$E = \frac{2\hbar^2 x^2}{m_w^* d^2}. \quad (6.28)$$

Our task is thus to solve eqn 6.26, with

$$\left( \frac{m_w^*}{m_b^*} \right)^{1/2} = \left( \frac{0.067}{0.092} \right)^{1/2} = 0.85,$$

and

$$\xi = \frac{0.067m_0 \times (10^{-8})^2 \times 0.30 \text{ eV}}{2\hbar^2} = 13.2.$$

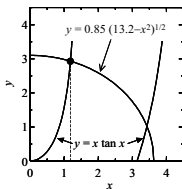


Fig. 6.5 Graphical solution of eqn 6.26 for the parameters given in Example 6.1.

Figure 6.5 plots the functions  $y = x \tan x$  and  $y = 0.85\sqrt{13.2 - x^2}$  on the same scales, and shows that the first value of  $x$  where the two functions are the same is  $x = 1.18$ . Hence from eqn 6.28 we find the required bound state energy:

$$E = \frac{2\hbar^2(1.18)^2}{0.067m_0 \times (10^{-8})^2} = 31.5 \text{ meV}.$$

This value of  $E$  can be compared to that given by eqn 6.13 for an infinite well:

$$E_1 = \frac{\hbar^2\pi^2(1)^2}{2 \times 0.067m_0 \times (10^{-8})^2} = 57 \text{ meV}.$$

The infinite well model thus overestimates the energy of the bound state by a factor of 1.8.

## 6.4 Quantum well absorption and excitons

In Sections 3.2 and 3.3 of Chapter 3 we used Fermi's golden rule to calculate the absorption spectrum of a bulk semiconductor. Then in Section 4.2 of Chapter 4 we studied how the spectrum is altered by excitonic effects. We now follow a similar approach for quantum wells, beginning with the selection rules and density of states for the optical transitions, and then moving on to consider the excitonic effects.

### 6.4.1 Selection rules

We consider a quantum well irradiated by light of angular frequency  $\omega$  propagating in the  $z$  direction, as shown in Fig. 6.6. The photons are absorbed by exciting electrons from an initial state  $|i\rangle$  at energy  $E_i$  in the valence band to a final state  $|f\rangle$  at energy  $E_f$  in the conduction band. Conservation of energy requires that  $E_f = (E_i + \hbar\omega)$ .

Fermi's golden rule tells us that the absorption rate is determined by the density of states and the square of the electric-dipole matrix element. (See Section B.2 in Appendix B.) The transition rate can be calculated by combining eqns 3.2, 3.3 and 3.6 to obtain:

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} | \langle f | -e\mathbf{r} \cdot \mathcal{E} | i \rangle |^2 g(\hbar\omega), \quad (6.29)$$

where  $\mathbf{r}$  is the position vector of the electron,  $\mathcal{E}$  is the electric field amplitude of the light wave, and  $g(\hbar\omega)$  is the density of states. We have simplified the form of the electric-dipole perturbation here by setting the  $e^{\pm i\mathbf{k} \cdot \mathbf{r}}$  factor of the light in eqn 3.6 equal to unity. As discussed in connection with eqn 3.12, this approximation is justified because the photon wave vector is negligible in comparison to that of the electron.

We first consider the matrix element for the transition. This will allow us to work out important selection rules. With photons incident in the

We are using Dirac notation here. See the marginal comment on p. 65.

$z$  direction as shown in Fig. 6.6, the polarization vector of the light is in the  $x, y$  plane. We therefore have to evaluate matrix elements of the form:

$$M = \langle f|x|i \rangle = \int \Psi_f^*(\mathbf{r}) x \Psi_i(\mathbf{r}) d^3\mathbf{r}. \quad (6.30)$$

When we considered matrix elements of this type in Section 3.2, it made no difference whether we evaluated  $\langle f|x|i \rangle$  or  $\langle f|y|i \rangle$  or  $\langle f|z|i \rangle$ . This was a consequence of the isotropy of the cubic semiconductors that we were considering. In the case of the quantum well, however, the  $x$  and  $y$  directions are equivalent, but the  $z$  direction is physically different. Therefore, for quantum wells we have:

$$\langle f|x|i \rangle = \langle f|y|i \rangle \neq \langle f|z|i \rangle. \quad (6.31)$$

In this section we concentrate on  $x, y$  polarized light, which is the usual experimental arrangement.

We are interested in evaluating eqn 6.30 for transitions between bound quantum well states in the valence and conduction bands. Figure 6.7 illustrates the type of transition we are considering. The figure specifically shows a transition from an  $n = 1$  hole level to an  $n = 1$  electron level, and from an  $n = 2$  hole level to an  $n = 2$  electron level.

We consider a general transition from the  $n$ th hole state to the  $n'$ th electron state. In analogy to the Bloch functions of eqns 3.7 and 3.8, we can use eqns 6.5 and 6.7 to write the initial and final quantum well wave functions in the form:

$$\Psi_i \equiv |i\rangle = \frac{1}{\sqrt{A}} u_v(\mathbf{r}) \varphi_{hn}(z) e^{i\mathbf{k}_{xy} \cdot \mathbf{r}_{xy}} \quad (6.32)$$

$$\Psi_f \equiv |f\rangle = \frac{1}{\sqrt{A}} u_c(\mathbf{r}) \varphi_{en'}(z) e^{i\mathbf{k}'_{xy} \cdot \mathbf{r}_{xy}}. \quad (6.33)$$

The three factors in these wave functions denote the envelope function for the valence or conduction band as appropriate, the bound states of the quantum well in the  $z$  direction, and the plane waves for the free motion in the  $x, y$  plane. We have written explicit subscripts to show that the plane waves only span the 2-D  $x, y$  coordinates.  $A$  is the normalization area in the  $x, y$  plane.

The momentum of the photon is very small in comparison to that of the electrons, and so conservation of momentum in the transition requires that  $\mathbf{k}_{xy} = \mathbf{k}'_{xy}$ . This is the 2-D equivalent of eqn 3.12 for 3-D bulk semiconductors. Therefore, on substituting eqns 6.32 and 6.33 into eqn 6.30, we see that the matrix element breaks into two factors:

$$M = M_{cv} M_{nn'} \quad (6.34)$$

where  $M_{cv}$  is the valence-conduction band dipole moment:

$$M_{cv} = \langle u_c|x|u_v \rangle = \int u_c^*(\mathbf{r}) x u_v(\mathbf{r}) d^3\mathbf{r}, \quad (6.35)$$

and  $M_{nn'}$  is the electron-hole overlap given by:

$$M_{nn'} = \langle en'|hn \rangle = \int_{-\infty}^{+\infty} \varphi_{en'}^*(z) \varphi_{hn}(z) dz. \quad (6.36)$$

We discuss the effect of having the light polarized in the  $z$  direction in Section 6.7.

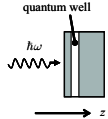
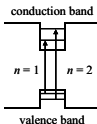


Fig. 6.6 Photons incident on a quantum well with light propagating in the  $z$  direction.



**Fig. 6.7** Interband optical transitions in a quantum well. The  $n = 1$  and  $n = 2$  transitions are indicated.

It will usually be the case that the constituent material of the quantum well (e.g. GaAs) has strongly allowed electric-dipole transitions between the conduction and valence bands. We considered this point in Section 3.3.1. Therefore, we can assume that  $M_{cv}$  is large and non-zero. Hence the matrix element for the optical transitions is proportional to the overlap of the electron and hole states given by eqn 6.36. This allows us to work out some straightforward selection rules on  $\Delta n = (n' - n)$ .

Consider first an infinite quantum well with wave functions of the form given by eqn 6.11. The overlap factor is:

$$M_{nn'} = \frac{2}{d} \int_{-d/2}^{+d/2} \sin\left(k_n z + \frac{n\pi}{2}\right) \sin\left(k_{n'} z + \frac{n'\pi}{2}\right) dz. \quad (6.37)$$

This is unity if  $n = n'$  and zero otherwise. (See Exercise 6.7.) Hence we obtain the following selection rule for an infinite quantum well:

$$\Delta n = 0. \quad (6.38)$$

This is why we only showed  $\Delta n = 0$  transitions in Fig. 6.7.

In finite quantum wells the electron and hole wave functions with differing quantum numbers are not necessarily orthogonal to each other because of their different decay constants in the barrier regions. This means that there are small departures from the selection rule of eqn 6.38. However these  $\Delta n \neq 0$  transitions are usually weak, and are strictly forbidden if  $\Delta n$  is an odd number, because the overlap of states with opposite parities is zero. (See Exercise 6.7.)

## 6.4.2 Two-dimensional absorption

The shape of the absorption spectrum in a quantum well can be understood by applying the selection rules we have just derived. If we increase the photon energy from zero, no transitions will be possible until we cross the threshold for exciting electrons from the ground state of the valence band (the  $n = 1$  heavy-hole level) to the lowest conduction band state (the  $n' = 1$  electron level). This is a  $\Delta n = 0$  transition and is therefore allowed. This threshold occurs at a photon energy given by

$$\hbar\omega = E_g + E_{hh1} + E_{e1}, \quad (6.39)$$

where  $E_g$  is the band gap of the quantum well material. This immediately gives us a very important result. The optical absorption edge of the quantum well has been shifted by  $(E_{hh1} + E_{e1})$  compared to the bulk semiconductor. Since the confinement energies can be varied by choice of the well width, this gives us a way to tune the frequency of the absorption edge.

The right-hand side of Fig. 6.8 shows the  $E-k_{xy}$  diagram for the transition between the  $n = 1$  levels. The bands have parabolic dispersions according to eqn 6.9. Conservation of momentum and the negligible  $\mathbf{k}$  vector of the photon imply that the electron and hole states have the

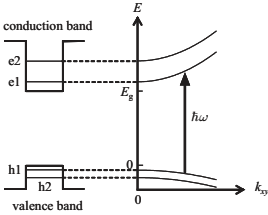


Fig. 6.8 The  $n = 1$  interband optical transition in a quantum well at finite  $k_{xy}$ .

same  $k_{xy}$  values. The energy of the transition shown by the vertical arrow is given by:

$$\begin{aligned}\hbar\omega &= E_g + \left( E_{hh1} + \frac{\hbar^2 k_{xy}^2}{2m_{hh}} \right) + \left( E_{e1} + \frac{\hbar^2 k_{xy}^2}{2m_e^*} \right) \\ &= E_g + E_{hh1} + E_{e1} + \frac{\hbar^2 k_{xy}^2}{2\mu},\end{aligned}\quad (6.40)$$

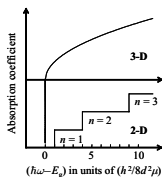
where  $\mu$  is the electron-hole reduced effective mass defined in eqn 3.22. This makes it clear that the transitions with  $\hbar\omega = (E_g + E_{hh1} + E_{e1})$  occur at  $k_{xy} = 0$ .

Equation 6.40 can be compared directly to eqn 3.23 for the bulk semiconductor. We have already noted the shift of the absorption threshold from  $E_g$  to  $(E_g + E_{hh1} + E_{e1})$ . The other crucial difference is that the wave vector in eqn 6.40 for the quantum well spans only the 2-D  $x, y$  coordinates, instead of the full 3-D  $x, y, z$  space. This has a very important consequence for the joint density of states factor that enters the transition rate in eqn 6.29. The 3-D bulk semiconductor had a parabolic density of states given by eqn 3.16, which led to the absorption edge given in eqn 3.25. By contrast, the joint density of states for a 2-D material is independent of energy and is given by (see Exercise 6.3):

$$g_{2D}(E) = \frac{\mu}{\pi\hbar^2}. \quad (6.41)$$

This means that the absorption coefficient will have a step-like structure, being zero up to the threshold energy given in eqn 6.39, and then having a constant non-zero value for larger photon energies.

The argument above can be repeated for the other allowed optical transitions in the quantum well. The next  $\Delta n = 0$  transition for the heavy-hole states occurs at an energy of  $(E_g + E_{hh2} + E_{e2})$  which corresponds to exciting an electron from the  $n = 2$  heavy-hole state to the  $n' = 2$  electron level. Once the photon energy crosses this threshold, the absorption coefficient will show a new step. There will also be other steps



**Fig. 6.9** The absorption coefficient for an infinite quantum well of width  $d$  compared to the equivalent bulk semiconductor.  $\mu$  is the electron-hole reduced mass. Excitonic effects are ignored.

corresponding to transitions from the light-hole states to the conduction band.

The functional form of the absorption coefficient for an infinite quantum well is shown in Fig. 6.9. The confinement energies of the electron and hole states are given by eqn 6.13, and the  $\Delta n = 0$  selection rule is strictly obeyed. The threshold energy for the  $n$ th transition is thus given by:

$$\hbar\omega = E_g + \frac{\hbar^2 n^2 \pi^2}{2m_e^* d^2} + \frac{\hbar^2 n^2 \pi^2}{2m_h^* d^2} = E_g + \frac{\hbar^2 n^2 \pi^2}{2\mu d^2}. \quad (6.42)$$

The spectrum therefore consists of a series of steps with threshold energies given by eqn 6.42. For comparison, the energy-dependence of the absorption coefficient for the equivalent bulk semiconductor is also plotted in Fig. 6.9. The shift of the absorption edge by the confinement energy is evident, together with the change of shape from the parabola of the bulk semiconductor to the step-like structure for the quantum well caused by the change in the density of states on going from 3-D to 2-D.

### Example 6.2

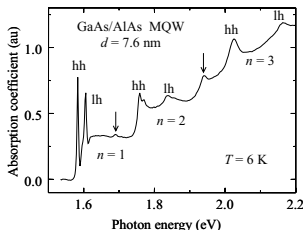
Estimate the difference in the wavelength of the absorption edge of a 20 nm GaAs quantum well and bulk GaAs at 300 K.

#### Solution

We see from eqn 6.39 that the absorption edge of a quantum well occurs at  $E_g + E_{hh1} + E_{e1}$ . We can estimate the confinement energies by using the infinite potential well model. By using eqn 6.13 and the effective mass data for GaAs given in Table D.2, we find that  $E_{hh1} = 2$  meV and  $E_{e1} = 14$  meV. These energies are small compared to typical quantum well barrier heights, and so the infinite well approximation should be reasonably accurate. The band edge therefore shifts from 1.424 eV to  $(1.424 + 0.002 + 0.014) = 1.440$  eV. This corresponds to a blue shift of 10 nm.

### 6.4.3 Experimental data

Figure 6.10 shows the absorption spectrum of a high-quality GaAs MQW structure containing 40 quantum wells of width 7.6 nm. The barriers were made of AlAs, and the sample temperature was 6 K. It is clear that the predicted step-like behaviour shown in Fig. 6.9 is well reproduced in the data, although the experimental spectrum is complicated by excitonic effects, which give rise to the strong peaks in the absorption at the edge of each step. These excitonic effects will be discussed further in Section 6.4.4, and we concentrate for now on the gross features of the absorption spectrum.



**Fig. 6.10** Absorption coefficient of a 40 period GaAs/AlAs MQW structure with 7.6 nm quantum wells at 6 K in arbitrary units. After Fox (1996), © Taylor & Francis Ltd, reprinted with permission.

The most pronounced steps in the spectrum are due to the  $\Delta n = 0$  transitions. The first of these occurs for the  $n = 1$  heavy-hole transition at 1.59 eV. This is closely followed by the step due to the  $n = 1$  light-hole transition at 1.61 eV. This should be compared with the low temperature band edge absorption spectra of bulk GaAs shown in Figs 4.3 and 4.4. We see that the band edge has been shifted in the quantum well by 0.07 eV.

The steps at the band edge are followed by a flat spectrum up to 1.74 eV in which the absorption is practically independent of energy. At 1.77 eV there is a further step in the spectrum due to the onset of the  $n = 2$  heavy-hole transition. This is followed by the step for the  $n = 2$  light-hole transition at 1.85 eV. Further steps due to the  $n = 3$  heavy- and light-hole transitions are observed at 2.03 eV and 2.16 eV respectively.

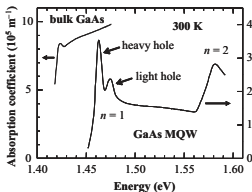
The two weak peaks identified by arrows are caused by parity-conserving  $\Delta n \neq 0$  transitions. The one at 1.69 eV is the  $hh3 \rightarrow e1$  transition, while that at 1.94 eV is the  $hh1 \rightarrow e3$  transition.

#### 6.4.4 Excitons in quantum wells

We now return to consider the excitonic effects that give rise to the sharp peaks that are very prominent in the experimental data shown in Fig. 6.10. As discussed in Chapter 4, excitons are bound electron-hole pairs held together by their mutual Coulomb attraction. Since the optical transition can be considered as the creation of an electron-hole pair, the Coulomb attraction increases the absorption rate because it enhances the probability of forming the electron-hole pair. Hence we observe peaks at the resonant energies for exciton formation. These peaks occur at the sum of the single particle energies less the binding energy of the bound pair. Detailed analysis of the data shown in Fig. 6.10 reveals that the binding energies of the quantum well excitons are about 10 meV. This is substantially higher than the value of 4.2 meV in bulk GaAs. (See Section 4.2.)

The enhancement of the excitonic binding energy in the quantum well

**Fig. 6.11** Absorption spectrum of a GaAs/Al<sub>0.28</sub>Ga<sub>0.72</sub>As MQW at room temperature. The structure contained 77 GaAs quantum wells of width 10 nm. The absorption spectrum of bulk GaAs at the same temperature is shown for comparison. After Miller et al. (1982), © American Institute of Physics, reprinted with permission.



The reason why the exciton binding energy is not enhanced by a factor of four is that a real quantum well is not a perfect 2-D system. The quantum well has a finite width, and the wave functions extend into the barriers due to tunnelling.

A striking difference between the quantum well and bulk absorption shown in Fig. 6.11 is the lifting of the degeneracy of the heavy- and light-hole states. The bulk sample shows a single excitonic shoulder at the band edge, but the quantum well shows two separate peaks. As discussed in point (2) of Section 6.3.2, this follows from the different effective masses of the heavy and light holes, and it highlights the lower symmetry of the quantum well sample.

is a consequence of the quantum confinement of the electrons and holes. This forces the electrons and holes to be closer together than they would be in a bulk semiconductor, and hence increases the attractive potential. It is possible to show that the binding energy of the ground-state exciton in an ideal 2-D system is enhanced by a factor of four compared to the bulk material. (See Exercise 6.10.) This should be compared with the factor of  $\sim 2.5$  deduced from the experimental data. Although we do not observe perfect 2-D enhancement of the binding energy, the increase is still substantial. The enhancement of the excitonic effects in quantum wells is very useful for device applications, as we discuss further in the next section.

One of the most useful consequences of the enhancement of the exciton binding energy in quantum wells is that the excitons are still stable at room temperature. This contrasts with bulk GaAs, which only shows strong excitonic effects at low temperatures. This can be clearly seen in the data shown in Fig. 6.11, which compares the absorption coefficient of a GaAs MQW structure with 10 nm quantum wells to that of bulk GaAs at room temperature. The bulk sample merely shows a weak shoulder at the band edge, but the MQW shows strong peaks for both the heavy-hole and the light-hole excitons. The more or less flat absorption coefficient expected for quantum wells above these peaks is also evident.

### 6.4.5 Spin injection in quantum wells

In Section 3.3.7 we studied how the excitation of a bulk semiconductor with circularly polarized light can lead to the generation of spin-polarized electrons. In the case of a III-V semiconductor with the zinc-blende structure such as GaAs, we found that the maximum spin polarization that can be generated is 50%. We now wish to reconsider this situation for the case of a quantum well. As we shall see, the lifting of the degeneracy of the heavy- and light-hole bands now makes it possible to generate fully spin-polarized electrons.

Figure 6.12 shows the optical transitions that can occur in a quantum well such as GaAs/AlGaAs in the presence of circularly polarized light.

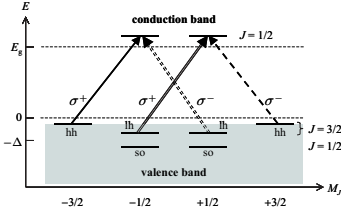


Fig. 6.12 Selection rules for circularly polarized light in a quantum well with the zinc-blende band structure. This diagram should be compared carefully to Fig. 3.8, which applies to bulk semiconductors. Note especially that the heavy- and light-hole transitions are no longer degenerate.

As discussed in Section 3.3.7, the valence-band states are derived from atomic  $p$  states split by the spin-orbit interaction, with the heavy- and light-hole levels corresponding respectively to the  $M_J = \pm 3/2$  and  $M_J = \pm 1/2$  sub-levels of the  $J = 3/2$  level. The conduction band has  $s$ -like atomic character, with  $M_J = \pm 1/2$  corresponding to spin up and down. The  $\sigma^\pm$  transitions have selection rules of  $\Delta M_J = \pm 1$  in absorption, which means that the transitions between the different sub-levels have the well defined circular polarizations indicated in Fig. 6.12.

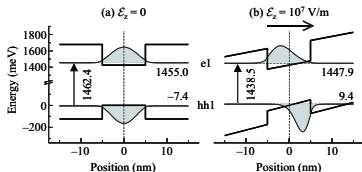
Figure 6.12 should be compared carefully to the equivalent one for bulk GaAs given in Fig. 3.8. The key difference is that the energies of the quantum well states are all shifted by the quantum confinement. The heavy- and light-hole levels are therefore no longer degenerate, and the energies of the  $hh1 \rightarrow e1$  and  $lh1 \rightarrow e1$  transitions are split by the difference in the  $hh1$  and  $lh1$  confinement energies. Hence by using circularly polarized light with energy in the range:

$$E_g + E_{e1} + E_{hh1} \leq \hbar\omega < E_g + E_{e1} + E_{lh1}, \quad (6.43)$$

it is possible to excite electrons with 100% spin polarization. The direction of the electron spin is determined by the polarization of the light, with  $\sigma^-$  or  $\sigma^+$  light corresponding respectively to spin-up or spin-down electrons.

The splitting of the heavy- and light-hole states caused by the quantum confinement has another important consequence. In a bulk semiconductor, the degeneracy of the heavy- and light-hole bands at  $k = 0$  means that the hole states have a mixed character, and therefore that the hole spin relaxation is very fast. This is no longer the case in a quantum well, where the  $M_J = \pm 3/2$  heavy-hole states are pure eigenstates, and circularly polarized photons with energy in the range set by eqn 6.43 create 100% spin-polarized heavy holes as well as electrons. When considering the spin relaxation processes in quantum wells, we therefore have to think about the holes as well as the electrons.

Fig. 6.13 Electron and heavy-hole wave functions for the first quantized levels of a 10 nm GaAs/Al<sub>0.3</sub>Ga<sub>0.7</sub>As quantum well at (a) zero field, and (b)  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$ . The energies are defined relative to the top of the valence band at the centre of the well (i.e. at  $z = 0$ ), and the band gap of GaAs is taken to be 1425 meV. In both diagrams the normalized probability densities (i.e.  $\psi^*\psi$ ) are plotted, and the numbers adjacent to the e1 and hh1 states give the level energies. The field direction in (b) is from left to right.



The mechanisms for spin relaxation in quantum wells are the same as those in bulk semiconductors, namely the EY, DP, and BAP processes. (See Section 5.3.4 for an explanation of these acronyms.) However, their relative importance can be affected by the quantum confinement. For example, the enhancement of excitonic effects in quantum wells and the fact that the heavy- and light-hole states are no longer degenerate means that we usually have to consider the spin relaxation of excitons rather than of individual electrons or holes. As a consequence, the BAP mechanism, which relies on electron-hole exchange, is more important in quantum wells than in the bulk. The reader is referred to the Further Reading list for a detailed discussion of the different spin relaxation processes that can occur.

## 6.5 The quantum-confined Stark effect

The quantum-confined Stark effect describes the response of a quantum-confined system to an external electric field. This makes it a type of electro-optic effect. As we discuss here, the response can be either linear or quadratic.

In Section 4.3.1 of Chapter 4 we considered the effects of a DC electric field on the excitons in bulk GaAs. We found that relatively small electric fields can ionize the excitons by pushing the electrons and holes in opposite directions. The situation in quantum wells is different if the field is applied along the  $z$  direction. The field still pushes the electrons and holes in opposite directions, but the barriers prevent the exciton from breaking apart. Hence the excitons are stable up to very high field strengths. These quantum-confined excitons interact with the field and shift to lower energy. In atomic physics, the shift of the energy levels in an electric field is called the Stark effect. The shift of the quantum-confined energy levels in a quantum well is therefore called the **quantum-confined Stark effect**.

When an electric field  $\mathcal{E}_z$  is applied to a semiconductor along the  $z$  axis, the potential energy of the electron is given by:

$$\Delta E = -p_z \mathcal{E}_z, \quad (6.44)$$

where  $p_z$  is the  $z$  component of the electron dipole. The electron is

negatively charged, and so we can write:

$$p_z = -ez, \quad (6.45)$$

where  $e$  is the modulus of the electron charge and  $z$  is the position along the  $z$  axis. The potential energy of the electron is then given by

$$\Delta E_e = +ez\mathcal{E}_z. \quad (6.46)$$

The application of the field therefore causes the potential energy of the electron to change linearly as a function of distance along the  $z$  axis.

Figure 6.13 illustrates the quantum-confined Stark effect for the first electron and heavy-hole states of a GaAs/Al<sub>0.3</sub>Ga<sub>0.7</sub>As quantum well with a thickness of 10 nm. Figure 6.13(a) shows the probability densities for the e1 and hh1 wave functions at zero field, while Fig. 6.13(b) shows the equivalent quantities at  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$ . In part (b) the linear increase of the valence and conduction band energy as a function of  $z$  is caused by adding a potential of the form given by eqn 6.46 to the potential well shown in part (a). The energies of the e1 and hh1 levels are indicated in both parts, with the zero of energy defined as the top of the valence band at the centre of the well (i.e. at  $z = 0$ ).

A number of important points are immediately obvious from comparison of Figs. 6.13(a) and (b).

- (1) For the electrons we have a confinement energy of  $(E_{e1} - E_s^{\text{GaAs}}) = (1455.0 - 1425) = 30.0 \text{ meV}$  at  $\mathcal{E}_z = 0$  and  $(1447.9 - 1425) = 22.9 \text{ meV}$  at  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$ , while for the holes the equivalent energies are  $7.4 \text{ meV}$  and  $-9.4 \text{ meV}$  respectively. The field thus causes a decrease of the energies of the confined states.
- (2) The Stark shift of the levels causes the energy difference between the e1 and hh1 levels (i.e.  $E_{e1} - E_{hh1}$ ) to decrease from  $1462.4 \text{ meV}$  at zero field to  $1438.7 \text{ meV}$  at  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$ . We thus expect a *red* shift of the transition energy.
- (3) At zero field the wave functions are symmetric about the centre of the well, but the application of the field breaks the symmetry and causes the electron and hole probability densities to shift in opposite directions.
- (4) The electron-hole overlap (see eqn 6.36) at zero field is nearly perfect with  $|\langle \varphi_{e1} | \varphi_{hh1} \rangle|^2 = 0.99$ . The skewing of the electron and hole wave functions in opposite directions at  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$  reduces the overlap to 0.38.

From these points it is apparent that the quantum-confined Stark effect is expected to cause a red shift of the lowest energy transition and a reduction in its oscillator strength.

The wave functions and level energies shown in Fig. 6.13 were calculated on a computer by numerical methods, which is the only option available for quantum wells with finite barriers. For infinite potential

Hole energies are measured *downwards* on electron band diagrams, and the confinement energy is therefore defined as  $-[E_{hh1} - E^v(z=0)]$ . The fact that this comes out to be negative at  $\mathcal{E}_z = 10^7 \text{ V m}^{-1}$  is not significant. The hole level actually lies  $40.6 \text{ meV}$  *above* the bottom of the well, which occurs at  $z = +5 \text{ nm}$ .

The e1-hh1 overlap is less than 100% at  $\mathcal{E}_z = 0$  because of the difference in the effective masses and barrier heights, which both affect the barrier penetration due to tunnelling. Perfect overlap is only expected for infinite barriers: see Exercise 6.7.

An exact solution exists for a particle confined in an infinite potential well with an electric field applied. See, for example, Miller (2008). Since this solution makes use of Airy functions, the mathematics is somewhat complicated, and is not discussed further here.

wells, however, analytic solutions exist. Perturbation theory gives the shift to the  $n = 1$  level in a small electric field as (see Exercise 6.14):

$$\Delta E = -24 \left( \frac{2}{3\pi} \right)^6 \frac{e^2 \mathcal{E}_z^2 m^* d^4}{\hbar^2}, \quad (6.47)$$

where  $d$  is the well width, and  $\mathcal{E}_z$  is the component of the field in the  $z$  direction. This result is analogous to the quadratic Stark effect in atomic hydrogen: the levels shift to lower energy in proportion to  $-\mathcal{E}_z^2$ .

The quadratic red shift of the levels can be understood as follows. The expectation value of the energy shift caused by the field is given by eqn 6.44 as:

$$\langle \Delta E \rangle = -\langle p_z \rangle \mathcal{E}_z, \quad (6.48)$$

where  $\langle p_z \rangle$  is the expectation value of the electron dipole. It follows from eqn 6.45 that  $\langle p_z \rangle$  is given by:

$$\langle p_z \rangle = -e \langle z_e \rangle, \quad (6.49)$$

where  $\langle z_e \rangle$  is the expectation value of the electron's position along the  $z$  axis:

$$\langle z_e \rangle = \int_{-\infty}^{+\infty} \varphi_e(z)^* z \varphi_e(z) dz. \quad (6.50)$$

At zero field, the  $n = 1$  electron wave function is symmetric about the centre of the well. This implies that both  $\langle z_e \rangle$  and  $\langle p_z \rangle$  are zero when no field is applied. With a finite field applied in the positive  $z$  direction, the electrons are pushed towards negative  $z$ , and  $\langle z_e \rangle$  acquires a negative value. This shifting of the average electron position in the opposite direction to the field is clearly shown by the  $e1$  wave function in Fig. 6.13(b). The shift of the electron creates a positive dipole which has a magnitude that is proportional to  $\mathcal{E}_z$  at small fields. Hence  $\langle p_z \rangle \propto +\mathcal{E}_z$  and therefore  $\langle \Delta E \rangle \propto -\mathcal{E}_z^2$ . The same argument can be applied to the holes.

We have seen in Section 6.4.4 that the absorption edge of a quantum well is actually dominated by excitonic effects. The shift of the exciton transition energy will be given by the sum of the shifts of the electron and hole levels, less any reduction in the exciton binding energy caused by the field. The latter effect is relatively small, and so the shift in the transition energy is well approximated by the sum of the shifts of the electron and hole levels:

$$\begin{aligned} \Delta(\hbar\omega) &= \langle \Delta E_e \rangle + \langle \Delta E_h \rangle, \\ &= -(\langle p_z^e \rangle + \langle p_z^h \rangle) \mathcal{E}_z, \\ &= -(-e \langle z_e \rangle + e \langle z_h \rangle) \mathcal{E}_z, \\ &= -e(\langle z_h \rangle - \langle z_e \rangle) \mathcal{E}_z. \end{aligned} \quad (6.51)$$

This shows that it is the displacement of the electron relative to the hole that causes the red shift of the transition. Furthermore, eqn 6.47 makes it clear that it is the hole that contributes the most to the energy shift, due to its larger effective mass. As discussed above, at small fields we would

expect the wave function displacements to be proportional to  $\mathcal{E}_z$ , giving rise to a quadratic Stark shift. However, this cannot continue indefinitely, because the displacement of the electron relative to the hole is limited by the width of the well. Hence the electron-hole dipole saturates at a value of order  $+ed$  at large fields, and the energy shift becomes linear in  $\mathcal{E}_z$ . There is some analogy here with the linear Stark effect of atomic physics, although the explanation is qualitatively different. In fact, Fig. 6.13(b) is close to the linear saturation limit, since a large field was deliberately chosen to exaggerate the electron-hole displacement.

The quantum-confined Stark effect can be observed by growing the quantum wells in the i-region of a p-i-n diode, as shown in Fig. 6.14. By operating the diode in reverse bias like a photodiode, strong DC electric fields can be applied in the growth direction. The magnitude of the field is given by eqn E.3 in Appendix E:

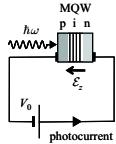
$$\mathcal{E}_z = \frac{|V_{bi} - V_0|}{l_i}, \quad (6.52)$$

where  $V_{bi}$  is the built-in voltage of the diode,  $V_0$  is the applied bias, and  $l_i$  is the i-region thickness.  $V_0$  is negative in reverse bias, and thus the applied voltage augments the field due to the built-in voltage. As discussed in Section 4.3.1, the photocurrent generated in the device follows the frequency dependence of the absorption.

Figure 6.15 shows the photocurrent spectra of a GaAs/Al<sub>0.3</sub>Ga<sub>0.7</sub>As MQW p-i-n diode at bias voltages of 0 V and -10 V. The well width was 9.0 nm, and the temperature was 300 K. The i-region thickness was 1  $\mu$ m, and  $V_{bi}$  was 1.5 V. From eqn 6.52, we see that the voltages correspond to field strengths of  $1.5 \times 10^6$  V m<sup>-1</sup> and  $1.15 \times 10^7$  V m<sup>-1</sup> respectively. The  $n = 1$  heavy- and light-hole exciton lines are clearly resolved at both field strengths, despite the fact that the temperature is 300 K and that  $\mathcal{E}_z$  considerably exceeds the exciton ionization field of bulk GaAs, which, as discussed in Section 4.3.1, is of order  $6 \times 10^5$  V m<sup>-1</sup>.

The spectrum at -10 V shows a clear red-shift for both the heavy- and light-hole excitons. As expected, the shift is larger for the heavy hole excitons due to their larger mass. The decrease in the exciton absorption at the higher field strength is caused by the reduction in the electron-hole overlap integral, as discussed in point (4) above. Two parity-forbidden transitions are clearly identified in Fig. 6.15, namely the hh2  $\rightarrow$  e1 and hh1  $\rightarrow$  e2 lines.

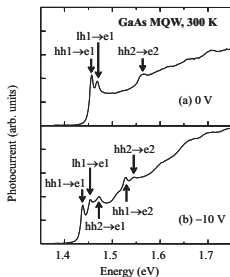
The ability to control the shape of the absorption spectrum by applying a voltage opens the possibility for making different types of electro-optic devices. The absorption at 1.44 eV (864 nm) in the quantum wells studied in Fig. 6.15 can be switched on and off by applying the bias. This allows us to make a voltage-tunable photodetector using the same arrangement as that shown in Fig. 6.14. The same device can also function as an intensity modulator by introducing a voltage-dependent loss on a 864 nm beam transmitted through the quantum wells. Moreover, the change of the absorption must also produce a change in the refractive index via the Kramers-Kronig relationships (see Section 2.3), and



**Fig. 6.14** Experimental set-up required to observe the quantum-confined Stark effect. The quantum wells are grown in the i-region of a p-i-n diode, and the device is operated in reverse bias. This produces a strong electric field  $\mathcal{E}_z$  across the quantum wells. The photocurrent generated when light is incident is determined by the absorption of the MQW layer.

The parity forbidden transitions with  $\Delta n$  equal to an odd number become allowed when the field is applied due to the lowering of the symmetry of the system. (See Exercise 6.17.)

**Fig. 6.15** Photocurrent spectra for a GaAs/Al<sub>0.3</sub>Ga<sub>0.7</sub>As MQW p-i-n diode with a 1  $\mu\text{m}$  thick i-region at room temperature. (a)  $V_0 = 0$ , (b)  $V_0 = -10$  V. The quantum well thickness was 9.0 nm. The transitions are labelled by the electron and hole states that participate. After Fox (1996), © Taylor & Francis Ltd, reprinted with permission.



so we can also use the device as a phase modulator.

## 6.6 Optical emission

The use of quantum well structures in electroluminescent devices is their main commercial application at present. By inserting quantum wells into the active region, a greater range of emission wavelengths can be obtained, together with an increase in the efficiency of the device.

The general principles of light emission in semiconductors were discussed in Chapter 5. The light is generated when electrons in the conduction band recombine with holes in the valence band. We saw in Sections 5.2–5.4 that the luminescence spectrum generally consists of a peak at the band gap energy with a width determined by the carrier density and the temperature.

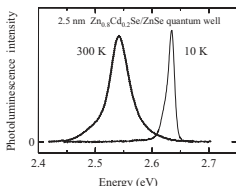
The physical processes responsible for light emission in quantum wells are essentially the same as those in bulk semiconductors. The electrons and holes injected electrically or optically rapidly relax to the bottom of their bands before emitting photons by radiative recombination. In a quantum well, the lowest levels available to the electrons and hole correspond to the  $n = 1$  confined states. Hence the low-intensity luminescence spectrum consists of a peak of spectral width  $\sim k_B T$  at energy

$$h\nu = E_g + E_{hh1} + E_{e1}. \quad (6.53)$$

This shows that the emission peak is shifted by the quantum confinement of the electrons and holes to higher energy compared to the bulk semiconductor.

Figure 6.16 shows the photoluminescence spectrum of a 2.5 nm ZnCdSe

The shape of the emission spectrum is only slightly affected by reducing the dimensionality of the system from 3-D to 2-D. The low-intensity emission spectrum of a bulk semiconductor is given by eqn 5.12. In a quantum well, the  $(h\nu - E_g)^{1/2}$  factor from the 3-D density of states will be replaced by the unit step function derived from the 2-D density of states. In both 3-D and 2-D the net result is that we get a peak of width  $\sim k_B T$  starting at the threshold energy for absorption.



**Fig. 6.16** Emission spectrum of a 2.5 nm  $\text{Zn}_{0.8}\text{Cd}_{0.2}\text{Se}/\text{ZnSe}$  quantum well at 10 K and at room temperature. The spectra have been normalized so that their peak heights are the same. Unpublished data from C.J. Stevens and R.A. Taylor.

quantum well with a cadmium concentration of 20%.  $\text{Zn}_{0.8}\text{Cd}_{0.2}\text{Se}$  is a II–VI alloy semiconductor with a direct band gap of 2.55 eV at 10 K, which corresponds to the blue/green spectral region. The barriers of the quantum wells are made from ZnSe, which has a band gap of 2.82 eV. The spectrum at 10 K peaks at 2.64 eV (470 nm) and has a full width at half maximum of 16 meV. The emission energy is about 0.1 eV larger than the band gap of the bulk material, and the linewidth is limited by the inevitable fluctuations in the well width that occur during the epitaxial growth. (See Exercise 6.18.) At room temperature the peak emission energy has shifted to 2.55 eV (486 nm) and the spectrum has broadened so that the linewidth is about 48 meV ( $\sim 2k_{\text{B}}T$ ).

As mentioned at the start of this section, the use of quantum wells in light-emitting devices is one of the main motivations for their development. Quantum wells offer three main advantages over the equivalent bulk materials:

- The shift of the luminescence peak by the confinement energy ( $E_{\text{e1}} + E_{\text{hh1}}$ ) allows the wavelength of light-emitting devices to be tuned by choice of the well width.
- The increased overlap between the electron and hole wave functions in the quantum well means that the emission probability is higher. This shortens the radiative lifetime, and the radiative recombination wins out over competing non-radiative decay mechanisms. The radiative efficiency is therefore higher in the quantum wells, which makes it easier to make bright light-emitting devices.
- The total thickness of the quantum wells in an electroluminescent device is very small ( $\sim 10$  nm). This is well below the critical thickness for dislocation formation in non-lattice-matched epitaxial layers. This allows the use of non lattice-matched combinations of materials, and hence gives even greater flexibility in emission wavelengths that can be obtained.

Electroluminescent devices can easily be made from quantum wells by incorporating them in the active region at the junction of a p–n diode, as discussed in Section 5.4 for bulk materials. The devices are operated

The shift to lower energy between 10 K and 300 K is caused by reduction of the ZnCdSe band gap with temperature. The emission energy is still about 0.1 eV above the band gap of the bulk  $\text{Zn}_{0.8}\text{Cd}_{0.2}\text{Se}$  at 300 K.

The blue and green emitting GaN-based structures grown on sapphire substrates mentioned in Section 5.4.1 are typical examples of a non-lattice-matched opto-electronic device. The commercial devices actually incorporate quantum wells in the active region. Other important examples include the 980 nm  $\text{Ga}_{1-x}\text{In}_x\text{As}$  quantum well lasers grown on GaAs substrates for use with optical fibre amplifiers: see Fig. 9.14 in Section 9.4.

in forward bias, and the light is emitted when the electrons and holes injected by the current recombine at the junction. GaAs quantum wells emitting around 800 nm are widely used as the lasers in compact disc players and printers. GaAs-based alloys are used to shift the wavelength into the red spectral region or into the infrared to match the optimal wavelengths for optical fibre systems at 1.3  $\mu\text{m}$  and 1.55  $\mu\text{m}$ . (See Exercise 6.19.)

### Example 6.3

Estimate the emission wavelength of a 15 nm GaAs quantum well laser at 300 K.

#### Solution

The emission wavelength is given by eqn 6.53. We estimate the confinement energies from eqn 6.13. Using the effective mass data given in Table D.2, we find  $E_{\text{eh}1} = 3 \text{ meV}$  and  $E_{\text{e}1} = 25 \text{ meV}$ . The emission energy is therefore  $1.424 + 0.003 + 0.025 = 1.452 \text{ eV}$ , which corresponds to a wavelength of 854 nm.

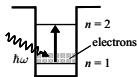


Fig. 6.17 The  $e1 \rightarrow e2$  intersubband transition in an n-doped quantum well.

## 6.7 Intersubband transitions

An **intersubband transition** is one in which we excite electrons and holes between the levels (or 'subbands') within the conduction or valence band. This contrasts with the interband transitions that we have been considering up till now in which the electrons move from the valence band to the conduction band and vice versa. Figure 6.17 illustrates a typical intersubband absorption transition in which an electron in the  $n = 1$  level of a quantum well is excited to the  $n = 2$  level by absorption of a photon.

A quick glance at Table 6.2 tells us that intersubband transitions occur at much lower photon energies than interband transitions. For example, the energy spacing between the  $n = 1$  and  $n = 2$  electron levels in a 10 nm GaAs/AlGaAs quantum well is of order 0.1 eV. This corresponds to an infrared wavelength of about 12  $\mu\text{m}$ . We can therefore use intersubband transitions to make detectors and emitters for the infrared spectral region by using GaAs quantum wells. This offers potential advantages over narrow gap semiconductors both in terms of performance and ease of fabrication.

Intersubband transitions are excited by light polarized along the  $z$  direction. The matrix element for such a transition from the  $n$ th to the  $n'$ th subband is  $\langle n|z|n' \rangle$ , and the selection rule on  $\Delta n = (n - n')$  is that  $\Delta n$  must be an odd number. (See Exercise 6.20.)

The requirement that the polarization must be along the  $z$  direction creates some technical difficulties. If the light is incident normal to the

surface as shown in Fig. 6.6, there is no polarization component along the  $z$  direction. To create a  $z$  component, the light must be incident at an angle. However, it is only possible to couple about 10% of the light to the intersubband transitions in this way, due to the high refractive index of semiconductors like GaAs. (See Exercise 6.21.) A better solution is to incorporate a metallic grating on the top of the sample. This can produce a substantial  $z$  component even for light incident normal to the surface.

Intersubband detectors have been under development for applications in infrared imaging since the late 1980s. They are made with  $n$ -doped quantum wells so that there is a large population of electrons in the  $n = 1$  level of the conduction band to absorb the light. In 1994 an intersubband laser called the ‘quantum cascade laser’ was reported. These intersubband lasers normally work at wavelengths in the infrared spectral region (i.e.  $\sim 5 - 15 \mu\text{m}$ ), and are required for many important applications such as atmospheric sensing. The lower wavelength limit is determined by the difference in the conduction band energies of the component semiconductors, while the upper wavelength limit is ultimately limited by the operating temperature, since the energy level spacing must satisfy  $E_2 - E_1 \gtrsim k_{\text{B}}T$  to prevent thermal occupation of the upper level. In 2002 an important breakthrough was achieved when the maximum wavelength for a quantum cascade laser was extended into the terahertz frequency range, that is  $\nu \lesssim 10 \text{ THz} \equiv \lambda \gtrsim 30 \mu\text{m}$ . Terahertz frequency sources are important for imaging and short-range wireless network applications.

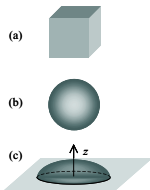
Experimental results show that the condition  $T \lesssim (E_2 - E_1)/k_{\text{B}}$  is actually too strict, and that quantum cascade lasers can operate at temperatures above this limit because they are non-equilibrium systems. Nevertheless, the highest operating temperature achieved at longer wavelengths does indeed scale with the laser frequency.

## 6.8 Quantum dots

We mentioned in Section 6.1 that other types of quantum-confined semiconductor structures can be made in addition to quantum wells. With reference to Table 6.1 and Fig. 6.1, we see that if we confine the electrons in all three directions, we have a quantum dot structure. With dimensions in the nanometre range, these quantum dots typically contain  $10^4 - 10^6$  atoms. It has been discovered that quantum dots can form spontaneously in some materials during epitaxial growth and colloidal synthesis. This makes it relatively easy to fabricate them, and accounts for the tremendous expansion in quantum dot research in recent years. Examples of these spontaneous dots will be given below, after first considering a few general principles of the physics of quantum dots.

### 6.8.1 Quantum dots as artificial atoms

In a quantum dot, the motion of the electrons and holes is confined in all three directions, so that it is appropriate to refer to a dot as a zero-dimensional structure. The electron and hole states are completely quantized, and the energy spectrum consists of a series of discrete lines, as illustrated schematically in Fig. 6.1. We have seen that the confinement of the electrons and holes in quantum wells gives rise to a host



**Fig. 6.18** Three different quantum dot shapes: (a) rectangular box, (b) sphere, (c) inverted lens.

of interesting optical effects and improvements in the performance of opto-electronic devices. It is thus to be expected that the increased degree of confinement in quantum dots should give further benefits. For example, by confining the carriers in all three dimensions, we increase the electron-hole overlap and thus increase the radiative quantum efficiency. Furthermore, the discrete nature of the density of states reduces the thermal spread of the carriers within their bands.

The task of calculating the level energies in a quantum dot is quite complicated, since many different shapes are possible. We consider below the shapes shown schematically in Fig. 6.18, namely the rectangular box, sphere, and inverted lens. These examples will serve to illustrate three important general points about quantum dots:

- (1) The energy spectrum and density of states are generically similar to those of an atom, with quantized states at discrete energies. Quantum dots are therefore sometimes described as ‘artificial atoms’.
- (2) The confinement energy scales as  $1/d^2$  where  $d$  is the size of the dot. This is consistent with the general argument about quantum confinement that leads to eqn 6.3.
- (3) The ground state is unique, but the excited states are degenerate. The ground state can therefore accommodate just two electrons (spin-up and spin-down) while the excited states can accommodate more.

### Rectangular box dots

The simplest shape to consider is that of a rectangular box with dimensions  $(d_x, d_y, d_z)$ , as shown in Fig. 6.18(a). In this case the problem separates, with the motion for each of the three axes given by a quantum well Schrödinger equation. If we assume that there are infinite potential barriers at the edges of the box, then the energy levels are given by (cf. eqn 6.13):

$$E(n_x, n_y, n_z) = \frac{\pi^2 \hbar^2}{2m^*} \left( \frac{n_x^2}{d_x^2} + \frac{n_y^2}{d_y^2} + \frac{n_z^2}{d_z^2} \right), \quad (6.54)$$

where the integer quantum numbers  $n_x$ ,  $n_y$  and  $n_z$  specify the quantized levels in each direction. The energy spectrum of a cubic dot with  $d_x = d_y = d_z$  is considered in Exercise 6.22. The ground state is unique, but the first and second excited states are both triply degenerate.

### Spherical dots

Spherical dots can be modelled by assuming that the electron experiences a potential of  $-V_0$  inside the dot and zero elsewhere. The spherical symmetry of the problem makes it convenient to work in spherical polar co-ordinates  $(r, \theta, \phi)$ . The Schrödinger equation for a dot of radius  $R_0$  is

then given by:

$$\left[ -\frac{\hbar^2}{2m^*} \nabla^2 + V(r) \right] \Psi(\mathbf{r}) = E \Psi(\mathbf{r}), \quad (6.55)$$

where  $V(r) = -V_0$  for  $r \leq R_0$ , and  $V(r) = 0$  for  $r > R_0$ . Since the potential only depends on  $r$ , the wave function can be separated into radial and angular components with:

$$\Psi(\mathbf{r}) = R_{nl}(r) Y_{lm}(\theta, \phi). \quad (6.56)$$

The angular wave functions  $Y_{lm}(\theta, \phi)$  are the spherical harmonic functions that appear in all 'central field' problems (i.e. those in which the force points along the radial direction from the origin, so that  $V(\mathbf{r})$  is a function of  $r$  only). The energy is found by solving the radial Schrödinger equation:

$$\left[ -\frac{\hbar^2}{2m^*} \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{d}{dr} \right) + \frac{\hbar^2 l(l+1)}{2m^* r^2} + V(r) \right] R_{nl}(r) = E_{nl} R_{nl}(r), \quad (6.57)$$

where  $n$  is an integer. The energy can in general be written in the form:

$$E_{nl} = \frac{\hbar^2}{2m^*} \frac{C_{nl}^2 \pi^2}{R_0^2}. \quad (6.58)$$

The values of  $C_{nl}$  for the first five bound states of an infinite well with  $V_0 = \infty$  are listed in Table 6.3. It is a straightforward exercise to show that  $C_{nl}$  is an integer when  $l = 0$ . (See Exercise 6.23.)

### Inverted lens dots

Quantum dots with the inverted lens shape shown in Fig. 6.18(c) are assumed to have cylindrical symmetry about the  $z$  (vertical) axis. This makes it convenient to work in cylindrical polar co-ordinates  $(r, \phi, z)$ , and to separate the lateral and vertical motions. It is further assumed that the size of the dots is much smaller in the  $z$  direction than in the lateral direction, and that the lateral potential approximates to a shallow harmonic oscillator with  $V(r, \phi) \propto r^2$  for small  $r$ . With these approximations, we may write the wave function in the form  $\Psi(\mathbf{r}) = \psi(r, \phi) \varphi(z)$ , and obtain separate Schrödinger equations for the lateral and vertical motions:

$$\left[ -\frac{\hbar^2}{2m^*} \nabla_{xy}^2 + \frac{1}{2} m^* \omega_0^2 r^2 \right] \psi(r, \phi) = E^{xy} \psi(r, \phi), \quad (6.59)$$

$$\left[ -\frac{\hbar^2}{2m^*} \frac{d^2}{dz^2} + V(z) \right] \varphi(z) = E^z \varphi(z), \quad (6.60)$$

with total energy given by:

$$E = E^{xy} + E^z. \quad (6.61)$$

Consider first the vertical ( $z$ ) motion. Equation 6.60 describes a potential well, which will have bound-state solutions with quantized energies.

The notation used to designate angular momentum states is summarized in Section C.2. The subscripts  $l$  and  $m$  refer to the orbital and magnetic quantum numbers respectively. In atomic physics, the properties of the Coulomb potential in the hydrogen atom require that  $l \leq (n-1)$ , where  $n$  is the principal quantum number. Since we do not have a Coulomb potential here, this restriction does not apply.

**Table 6.3** First five bound states of a spherical potential well with an infinite barrier.  $g$  denotes the degeneracy including the spin. After Bimberg et al. (1999).

| Level       | $n$ | $l$ | $g$ | $C_{nl}$ |
|-------------|-----|-----|-----|----------|
| Ground      | 1   | 0   | 2   | 1        |
| 1st excited | 1   | 1   | 6   | 1.43     |
| 2nd excited | 1   | 2   | 10  | 1.83     |
| 3rd excited | 2   | 0   | 2   | 2        |
| 4th excited | 1   | 3   | 14  | 2.22     |

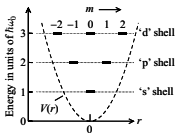


Fig. 6.19 First three energy levels of a two-dimensional harmonic oscillator. The allowed values of  $m$  are indicated, together with the functional form of the potential.

Since the dimensions are very small, it is likely that there will only be one bound state, or if there are more, that the energy difference to the second level will be much larger than the energy separation of the levels associated with the lateral motion. It is therefore reasonable to assume that we only need to consider one bound state for the vertical motion with energy  $E_1^z$ .

Now consider the lateral motion. The Schrödinger equation of eqn 6.59 describes a two-dimensional harmonic oscillator. By reverting to Cartesian co-ordinates, it is easy to show that the energy is given by:

$$E^{xy} = (n+1)\hbar\omega_0, \quad (6.62)$$

where  $n = 0, 1, 2, \dots$  (See Exercise 6.25.) The total energy is therefore given by:

$$E_n = E_1^z + (n+1)\hbar\omega_0. \quad (6.63)$$

The  $n$ th level has  $n$  degenerate sub-levels which are identified by the quantum number  $m$ . This quantifies the angular momentum about the  $z$  axis, and can take integer values from  $-n$  to  $+n$  in steps of two. The allowed values of  $m$  for the first three levels are shown in Fig. 6.19. In analogy with atomic physics, the levels are often labelled as 's', 'p', 'd', etc., shells, although the analogy is not exact, since the notation originates from the analysis of 3-D central fields.

The arguments above can be generalized to dots with more complicated shapes but which still satisfy the criterion that the vertical dimensions are significantly smaller than the lateral ones, and that the lateral confining potential is relatively weak. In this case it will normally be the case that we only need to consider the first bound state for the vertical motion, and the first few excited states arise from the lateral motion. When the cylindrical symmetry about the  $z$  axis is lost (e.g. if the dot has an elliptical shape in the  $x$ - $y$  plane), we would no longer expect the sub-levels of the  $n$ th level to be degenerate.

## 6.8.2 Colloidal quantum dots

One of the ways that quantum dots can be made is by techniques of colloidal synthesis. The resulting dots, which are commonly made from II-VI or III-V semiconductors, are spherical in shape, and can be studied in solution or in the solid state. In the latter case, the dots are either deposited as thin films on transparent substrates or are doped into a glass.

Figure 6.20 shows the absorption spectra of two different types of II-VI colloidal dots. Figure 6.20(a) shows the data for CdSe dots deposited on sapphire substrates at 10 K, while Fig. 6.20(b) shows that for CdTe dots in solution at room temperature. In both cases, the spectra for several different sizes of dots are shown. The diameters quoted represent the average values. In practice there is always a spread of sizes, and this causes broadening of the optical spectra. In fact, a large part of the art of making colloidal quantum dots is precisely in obtaining good size

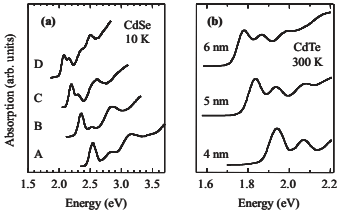


Fig. 6.20 (a) Absorption spectra of close-packed CdSe colloidal quantum dots deposited on sapphire substrates at 10 K. The dot diameters were: (A) 3.03 nm, (B) 3.94 nm, (C) 4.80 nm, and (D) 6.21 nm. (b) Close-up of the band edge of CdTe colloidal quantum dots in solution at room temperature. After Kagan et al. (1996), © American Physical Society, and De Giorgi et al. (2005), © Elsevier, reprinted with permission.

control. For example, the standard deviation of the diameters of the dots shown in Fig. 6.20(a) was  $< 5\%$ .

In both sets of spectra in Fig. 6.20, there is a clear size-dependent shift of the absorption edge relative to the band gap. ( $E_g = 1.85$  eV for CdSe at 10 K and 1.50 eV for CdTe at room temperature.) Excitonic peaks associated with the confined electron and hole states are clearly resolved at both temperatures. The absorption edge occurs at the energy of the exciton from the first confined state of the electrons and holes. If we make the assumption that  $C_{10} \approx 1$  for both the electrons and holes, then eqn 6.58 implies that the band edge should occur at:

$$\hbar\omega \approx E_g + \frac{2\pi^2\hbar^2}{\mu d^2} - E_X, \quad (6.64)$$

where  $\mu$  is the electron-hole reduced mass,  $d$  is the dot diameter, and  $E_X$  is the exciton binding energy. The shift of the absorption edge in proportion to  $1/d^2$  is a very clear demonstration of the quantum size effects that we have been discussing throughout this chapter. Note that the magnitude of the quantum size effect is very large, with a shift of over 0.6 eV for the CdSe sample with  $d = 3.03$  nm.

One application of colloidal dots is in semiconductor-doped glasses. As mentioned in Section 1.4.5, II–VI semiconductors such as CdS, CdSe, ZnS, and ZnSe can be introduced into a glass during the melt process, and colloidal quantum dots can be formed in the right conditions. The size of the dots depends on the way the glass is produced, and good uniformity can be achieved with careful preparation. The size-dependent shift of the absorption shown in Fig. 6.20 provides a way to control the colour of filters made from semiconductor-doped glass. In this way it is possible to make colour-glass filters spanning most of the visible spectral region just by altering the size of the dots.

Another important application of colloidal quantum dots is in fluorescence imaging in chemistry and biology. The dots are prepared in solution and are attached to the molecules under study by synthetic

techniques. By exploiting their high luminescence efficiency and tunable emission energy, the dots can act as ‘marker tags’ to identify the molecule under optical illumination. Organic dyes were formerly used for this application, but quantum dots give sharper lines, and are also more convenient to use.

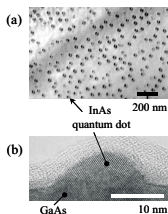
### 6.8.3 Self-assembled epitaxial quantum dots

In the 1990s it was discovered that quantum dots would spontaneously form during epitaxial crystal growth in the Stranski–Krastanow regime. Many different types of III–V and II–VI quantum dots have now been made in this way by using MBE or MOCVD. The fact that the dots can be formed directly during epitaxial growth means that they can easily be incorporated into laser diode structures, and the 0-D physics causes a reduction in both the magnitude of the threshold current and its sensitivity to the temperature.

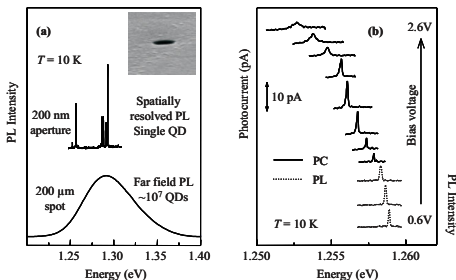
The most common example of Stranski–Krastanow dots is InAs in GaAs. InAs is a narrow gap semiconductor with a unit cell size that is 7% larger than that of GaAs. The dots are formed when thin layers of InAs molecules are deposited on GaAs during MBE growth. The InAs molecules try to adopt the lattice constant of the GaAs, and this leads to the formation of a highly strained layer called the ‘wetting layer’ on the surface of the crystal. The energy required to strain the layer is so large that it ceases to be favourable to form a uniform layer when the thickness exceeds only a few atoms. Instead, the InAs molecules coalesce into clusters, leading to the formation of InAs quantum dots on top of the wetting layer. By depositing layers of GaAs on top of the dots, the electrons and holes are then confined in both the vertical and lateral directions.

Figure 6.21 shows transmission electron microscope (TEM) images of InAs quantum dots grown by the Stranski–Krastanow technique. Part (a) shows a plan view, while part (b) shows a side view looking down the wafer edge at higher resolution. The TEM images show that the lateral and vertical dimensions of the dot are both in the nanometre range, leading to strong confinement in all directions. It is apparent from Figs 6.21(a) and (b) that the shape of the dots approximates to the inverted lens modelled in Section 6.8.1. The uniformity of the size distribution is apparent in Fig. 6.21(a): most of the dots are roughly of the same size, but some variation is clearly visible.

The appropriateness of describing quantum dots as ‘artificial atoms’ becomes most clear when techniques are used to isolate the optical spectra of individual dots. Figure 6.22 illustrates typical spectra than can be observed in this way. The lower spectrum in Fig. 6.22(a) shows the photoluminescence spectrum obtained when an InAs/GaAs quantum dot wafer is excited by a laser with a large spot size. Several million dots fall within the illuminated area, and this results in a broad spectrum that reflects the size distribution of the dots. Note that the quantum confinement increases the emission energy by almost 1 eV from the band gap



**Fig. 6.21** (a) Plan view of an uncapped layer of InAs quantum dots formed during Stranski–Krastanow growth on a GaAs crystal. (b) Side image of one of the InAs dots looking down the edge of the wafer. The mottled pattern above the dot originates from the adhesive used to hold the sample in position. Both images were taken with a transmission electron microscope. After M. Hopkinson (unpublished), and Fry et al. (2000), © American Physical Society, reprinted with permission.



**Fig. 6.22** (a) Photoluminescence (PL) spectra of InAs/GaAs self-organized quantum dots at 10 K. The spectrum obtained for a large area laser spot is compared to that obtained through a nano-aperture similar to the one shown in the inset. (b) Voltage dependence of the PL and photocurrent (PC) spectra of a single InGaAs/GaAs quantum dot incorporated into a Schottky diode at 10 K. The spectra were observed through a nano-aperture of diameter 800 nm, and are given for bias voltages from 0.6 V to 2.6 V in steps of 0.2 V. After J.J. Finley & M.S. Skolnick (unpublished), and Oulton et al. (2002), © American Physical Society, reprinted with permission.

of bulk InAs (0.4 eV). The upper spectrum in Fig. 6.22(a) illustrates the results obtained when exciting the dots through a nano-aperture of diameter 200 nm. This size of the aperture is chosen so that the light from only a few dots, at most, is collected. The spectrum breaks up into a series of sharp emission lines from the ground- and excited-state excitons of individual quantum dots. If there is more than one dot under the aperture, the change of the dot energy due to inevitable size and shape variations allows the lines of individual dots to be identified by spectral selection.

The width of the exciton line from a single quantum dot is ultimately limited by the radiative lifetime. With typical values of  $\tau_R$  being around 1 ns, linewidths as small as a few  $\mu$ eV have been observed. Since a single quantum dot can only emit one photon at a time on a particular transition, it can act as a **single-photon source**, with many potential applications in quantum optics.

Figure 6.22(b) shows the voltage dependence of the excitonic spectra from a single InGaAs/GaAs quantum dot embedded within a Schottky diode at 10 K. A combination of photoluminescence (PL) and photocurrent (PC) techniques were used, with a tunable Ti:sapphire as the excitation source. For the PL spectra, the excitation energy was set at  $\sim 1.35$  eV (i.e. well above the exciton energy), while for the PC measurements, the laser was tuned through the exciton line. The bias voltage

Some of the lines observed in the spectrum of the single dot might arise from charged excitons. For example, the dot might trap a free electron from the conduction band, and then a negatively charged exciton containing two electrons and one hole can be formed when an electron-hole pair is excited by a photon. The energies of these charged excitons are slightly shifted from those of the neutral ones by Coulomb interactions.

controls the electric field across the quantum dot. At low bias voltages, the field is small, and the excitons recombine by emitting photons. As the bias voltage is increased, the field increases, which leads to an increased probability that the electrons and holes tunnel out of the dot before recombining. This causes a reduction of the luminescence efficiency, with a concomitant increase in the photocurrent efficiency. At the same time, the exciton shifts to lower energy due to the quantum-confined Stark effect, and the line strength decreases due to the field-induced reduction in the electron-hole overlap.

All of these effects are clearly observed in Fig. 6.22(b). The most obvious feature of the data is the red shift with increasing voltage due to the quantum-confined Stark effect. The PL efficiency is high at 0.6 V but then drops rapidly above 1.0 V. At the same time, the PC efficiency rises above 1.0 V, reaching its maximum value (assumed to be unity) at 1.8 V. The line intensity then decreases again as the electron-hole overlap decreases, and broadening is observed as the lifetime of the exciton shortens due to tunnelling of electrons and holes out of the dot. These results are typical for many other types of III-V and II-VI quantum dots.

## Chapter summary

- Quantum confinement occurs when the dimensions of the structure are small enough that the confinement energy is greater than the thermal energy at that temperature.
- A structure with confinement in one dimension is called a quantum well. Structures with confinement in two or three dimension are called quantum wires and quantum dots respectively.
- Semiconductor quantum wells are made by epitaxial growth of very thin layers. The quantum confinement arises from the potential barriers at the interfaces between different semiconductors due to their different band gaps. The electrons and holes exhibit two-dimensional physics.
- A multiple quantum well is a crystal containing many quantum wells that are separated from each other. A superlattice is a similar structure but with thinner barriers, so that adjacent wells are coupled together by tunnelling through the barriers.
- The energies of the confined states in a quantum well can be calculated by modelling the system as a one-dimensional potential well with a depth determined by the difference in the band gaps of the constituent semiconductors.

- The quantum confinement shifts the absorption edge of a quantum well to higher energy compared to the bulk semiconductor. The absorption spectrum is mainly determined by the 2-D density of states of the quantum well and consists of a series of steps. The splitting of the heavy- and light-hole bands allows the generation of 100% spin polarized electrons by absorption of circularly polarized light.
- Excitonic effects are enhanced in quantum wells. Exciton absorption peaks are readily observed at room temperature in the absorption spectra.
- The quantum-confined Stark effect is the shift of the quantum well levels induced by an electric field, which causes a red shift in the band edge and exciton energies. The effect can be used to make optical modulators.
- The emission energy for luminescence is larger than in a bulk semiconductor due to the quantum confinement of the electrons and holes. Quantum wells make bright light-emitting devices, and the emission wavelength can be tuned by choice of the quantum well parameters.
- Intersubband transitions occur when electrons are excited between the subbands of a quantum well by absorption of a photon. The transitions occur at infrared wavelengths.
- Quantum dots exhibit zero-dimensional physics, with fully quantized energies and a discrete density of states. The dots can be made by colloidal synthesis or by self-organization during epitaxial growth in the Stranski–Krastanow regime. The absorption and emission spectra show broad lines due to variations in the dot sizes, but very sharp lines can be observed by isolating the spectra of individual quantum dots by nano-aperture techniques.

## Further reading

The seminal paper on semiconductor quantum wells is Esaki and Tsu (1970). Complementary introductory reading to the treatment given here may be found in Burns (1985) or Singleton (2001). The subject is treated at a more thorough level in Yu and Cardona (1996), or in a number of specific quantum well texts, such as Bastard (1990), Jaros (1989), Kelly (1995), Singh (1993), or Weisbuch and Vinter (1991).

The spin dynamics of electrons, holes, and excitons in quantum wells is discussed in the collection of articles in Awschalom et al. (2002) and Dyakonov (2008). A specific review of the subject may be found in Viña (1999).

The Stark effect in hydrogen is covered in most quan-

tum mechanics texts, for example Gasiorowicz (1996) or Schiff (1969), and also in atomic physics texts, such as Woodgate (1980). A thorough discussion of the effects of electric fields on particles confined in potential wells may be found in Miller (2008), while Chuang (1995) covers the physics of the quantum-confined Stark effect in depth.

Blood (1999) gives a review of the use of quantum wells in visible-emitting diode lasers. The physics of intersubband transitions is discussed in Helm (2000), Liu and Capasso (2000a), and Liu and Capasso (2000b). The terahertz quantum cascade laser is reviewed in Williams (2007), and a collection of articles describing the significance of terahertz technology may be found in Davies et

al. (2004).

The physics of quantum dots is described by Bimberg et al. (1999), Harrison (2005), and Woggon (1997). Murray et al. (2000) give a detailed description of the synthe-

sis and characterization of colloidal quantum dots, while a discussion of the spectra of single quantum dots and their application in quantum optics may be found in Michler (2003).

## Exercises

- (6.1) Estimate the temperature at which quantum size effects would be important for a semiconductor layer of thickness  $1 \mu\text{m}$  if the effective mass of the electrons is  $0.1m_0$ .
- (6.2) A particle of mass  $m^*$  is confined to move in a quantum well of width  $d$  with infinite barriers. Show that the energy separation of the first two levels is equal to  $k_B T/2$  when  $d$  is equal to  $d = (3\hbar^2/4m^*k_B T)^{1/2}$ . Evaluate  $d$  for electrons of effective mass  $m_0$  and  $0.1m_0$  at 300 K. Hence show that this value of  $d$  is smaller than the value of  $\Delta x$  given in eqn 6.4 by a factor of  $\sqrt{3\pi^2}$ .
- (6.3) Consider a gas of spin  $1/2$  particles of mass  $m$  moving in a two-dimensional layer. Apply Born-von Karman periodic boundary conditions (i.e.  $e^{ikx} = e^{ik(x+L)}$ , etc, where  $L$  is a macroscopic length) to show that the density of states in  $k$ -space is  $1/(2\pi)^2$ . By considering the incremental area enclosed by two circles in  $k$ -space differing in radius by  $dk$ , show that the number of states with  $k$  vectors between  $k$  and  $k+dk$  is given by  $g(k)dk = (k/2\pi)dk$ . Hence show that if the energy dispersion is given by  $E(k) = \hbar^2 k^2/2m$ , the density of states in energy space is given by

$$g_{2D}(E) dE = \frac{m}{\pi \hbar^2} dE.$$

- (6.4) Repeat Exercise 6.3 for a gas of free spin  $1/2$  particles of mass  $m$  moving in a one-dimensional wire to show that the density of states in energy space is given by:

$$g_{1D}(E) dE = \sqrt{m/2\hbar^2\pi^2} E^{-1/2} dE.$$

- (6.5) Explain, with reference to eqn 6.26, why a finite quantum well will always have at least one bound state, no matter how small  $V_0$  is.
- (6.6) Calculate the energy of the first heavy-hole bound state of a GaAs/AlGaAs quantum well with  $d =$

$10 \text{ nm}$ . Take  $V_0 = 0.15 \text{ eV}$ ,  $m_w^* = 0.34m_0$  and  $m_h^* = 0.5m_0$ . How does this energy compare to that of an equivalent well with infinite barriers?

- (6.7) Consider the electron-hole overlap integral  $M_{nn'}$  for a quantum well given by:

$$M_{nn'} = \int_{-\infty}^{+\infty} \varphi_{en'}^*(z) \varphi_{hn}(z) dz.$$

- (a) Show that  $M_{nn'}^*$  is unity if  $n = n'$  and zero otherwise in a quantum well with infinite barriers.  
 (b) Show that  $M_{nn'}$  is zero if  $(n - n')$  is an odd number in a quantum well with finite barriers.
- (6.8) Draw a sketch of the energy dependence of the absorption spectrum of a  $5 \text{ nm}$  GaAs quantum well at 300 K between  $1.4 \text{ eV}$  and  $2.0 \text{ eV}$ . Assume that the confining barriers are infinite and ignore excitonic effects. See Table D.2 for band structure data on GaAs.
- (6.9) Discuss how the spectrum in Exercise 6.8 would change if (a) the barrier height were finite, and (b) excitonic effects were included.
- (6.10)\* The variational technique introduced in Exercise 4.4 can be used to calculate the energy and radius of a 2-D exciton.<sup>1</sup> The Hamiltonian for the relative motion of an electron-hole pair in a 2-D material is given in polar coordinates by

$$\hat{H} = -\frac{\hbar^2}{2\mu} \left( \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} \right) - \frac{e^2}{4\pi\epsilon_0\epsilon_r r},$$

where  $r^2 = (x^2 + y^2)$ . As for the 3-D exciton considered in Exercise 4.4, we guess a trial wave function with a 1s-like radial dependence:

$$\Psi(r, \phi) = \left( \frac{2}{\pi \xi^2} \right)^{1/2} \exp \left( -\frac{r}{\xi} \right),$$

where  $\xi$  is the variational parameter.

- (a) Verify that the trial wave function is properly

\*Exercises marked with an asterisk are more difficult.

<sup>1</sup>The 2-D exciton problem can also be solved exactly using Laguerre polynomials, but the variational approach is easier.

normalized.

(b) The variational energy  $\langle E \rangle_{\text{var}}$  is given by:

$$\langle E \rangle_{\text{var}} = \int_{r=0}^{\infty} \int_{\phi=0}^{2\pi} \Psi^* \hat{H} \Psi r dr d\phi.$$

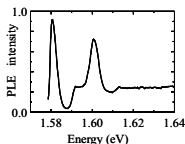
Show that  $\langle E \rangle_{\text{var}}$  is given by:

$$\langle E \rangle_{\text{var}} = \frac{\hbar^2}{2\mu\xi^2} - \frac{e^2}{2\pi\epsilon_r\epsilon_0\xi}.$$

(c) Vary  $\langle E \rangle_{\text{var}}$  with respect to  $\xi$  to obtain the best estimate for the energy. Show that this is four times larger than that of the equivalent bulk semiconductor.

(d) Show the Bohr radius of the 2-D exciton, namely the value of  $\xi$  that minimizes  $\langle E \rangle_{\text{var}}$ , is half that of the equivalent 3-D exciton.

- (6.11)\* Discuss qualitatively how you would expect the exciton binding energy in a GaAs/Al<sub>0.3</sub>Ga<sub>0.7</sub>As quantum well to vary with the quantum well thickness, given that the binding energy of the excitons in bulk GaAs and Al<sub>0.3</sub>Ga<sub>0.7</sub>As are 4 meV and 6 meV respectively.



**Fig. 6.23** Photoluminescence excitation (PLE) spectrum of a GaAs/AlAs quantum well at 4 K. Unpublished data from R.A. Taylor.

- (6.12) Figure 6.23 shows the absorption spectrum of a GaAs/AlAs quantum well at 4 K measured by using the photoluminescence excitation technique. Relevant band structure data for GaAs is given in Table D.2.

(a) Explain the principles of photoluminescence excitation spectroscopy.  
 (b) Account for the shape of the absorption spectrum.  
 (c) Estimate the width of the quantum wells by assuming that they behave like a perfect 2-D system with infinite barriers. Would you expect the

true well width to be larger or smaller than the answer you have worked out this way?

(d) Deduce the binding energies of the heavy- and light-hole excitons, and comment on the values you obtain.

- (6.13) Discuss the variation of the electron spin polarization created with  $\sigma^+$  light in a GaAs/AlGaAs quantum well as the photon energy is varied above the fundamental absorption edge.

- (6.14)\* The Stark shift of the confined levels in a quantum well can be calculated by using second-order perturbation theory. Consider the interaction between the electrons in a quantum well of width  $d$  and a DC electric field of strength  $\mathcal{E}_z$  applied along the  $z$  (growth) axis. Equation 6.46 shows that the perturbation is of the form  $H' = e\mathcal{E}_z z$ .

(a) Explain why the first-order shift of the energy levels, given by:

$$\Delta E^{(1)} = \int_{-\infty}^{+\infty} \varphi(z)^* H' \varphi(z) dz,$$

is zero.

(b) The second-order energy shift of the  $n = 1$  level is given by:

$$\Delta E^{(2)} = \sum_{n>1} \frac{|\langle 1|H'|n\rangle|^2}{E_1 - E_n},$$

where

$$\langle 1|H'|n\rangle = \int_{-\infty}^{+\infty} \varphi_1(z)^* H' \varphi_n(z) dz.$$

This can be evaluated exactly if we have infinite confining barriers. Within this approximation, show that the Stark shift is given approximately by:

$$\Delta E = -24 \left( \frac{2}{3\pi} \right)^6 \frac{e^2 \mathcal{E}_z^2 m^* d^4}{\hbar^2}.$$

- (6.15) The magnitude of the  $n = 1$  heavy-hole exciton red shift is 10.5 nm at  $-10$  V for the MQW diode studied in Fig. 6.15. The p-i-n sample has an i-region thickness of 1.0  $\mu\text{m}$  and a built-in voltage of 1.5 V.

(a) Compare the magnitude of the Stark shift to that predicted in Fig. 6.13, and account for any difference.  
 (b) Estimate the magnitude of the red shift at  $-5$  V.  
 (c) Estimate the average relative displacement of the electron and hole probability densities at  $-10$  V.

**Table 6.4** Dependence of the  $n = 1$  heavy-hole transition on the electric field strength  $\mathcal{E}_z$  for two GaAs quantum well samples. Sample A had a well width of 10 nm, while sample B had a well width of 18 nm. The transition energies are given in eV.

| $\mathcal{E}_z$ (Vm <sup>-1</sup> ) | A     | B     |
|-------------------------------------|-------|-------|
| 0                                   | 1.548 | 1.524 |
| $3 \times 10^6$                     | 1.547 | 1.518 |
| $6 \times 10^6$                     | 1.543 | 1.497 |
| $9 \times 10^6$                     | 1.535 | 1.470 |

- (6.16) Table 6.4 gives experimental data for the red shift of the  $n = 1$  heavy-hole transition due to the quantum-confined Stark effect in two GaAs quantum well samples with well widths of 10 nm and 18 nm respectively. Compare the experimental data with the predictions of Exercise 6.14(b), and account qualitatively for any major discrepancies. Band structure data for GaAs is given in Table D.2.

- (6.17) A DC electric field of magnitude  $\mathcal{E}_z$  is applied along the growth ( $z$ ) axis of a quantum well. Use symmetry arguments to explain why transitions between confined electron and holes states with  $\Delta n$  equal to an odd number are forbidden at  $\mathcal{E}_z = 0$ , but not at finite  $\mathcal{E}_z$ .

- (6.18) By assuming that the confinement energy varies as  $d^{-2}$ , estimate the shift in the luminescence emission energy caused by a  $\pm 5\%$  change in  $d$  for a 2.5 nm ZnCdSe quantum well, given that the total confinement energy for the electrons and holes at  $d = 2.5$  nm is 0.1 eV. Compare this value to the measured linewidth of the 10 K data shown in Fig. 6.16, and comment on the answer. The unit cell size of the crystal is 0.28 nm, and the electron and hole effective masses of Zn<sub>0.8</sub>Cd<sub>0.2</sub>Se are  $0.15m_0$  and  $0.5m_0$  respectively.

- (6.19) A Ga<sub>0.47</sub>In<sub>0.53</sub>As quantum well laser is designed to emit at 1.55  $\mu$ m at room temperature. Estimate the width of the quantum wells within the device. ( $E_g = 0.75$  eV,  $m_e^* = 0.041m_0$ ,  $m_{hh}^* = 0.47m_0$ .)

- (6.20)\* The matrix element for an intersubband transition between the  $n$ th and  $n'$ th subbands of a quantum well is given by:

$$\langle n|z|n' \rangle = \int_{-\infty}^{+\infty} \varphi_n^*(z) z \varphi_{n'}(z) dz.$$

- (a) By considering the parity of the states, prove that  $\Delta n = (n - n')$  must be an odd number.  
 (b) Compare the relative strengths of the  $1 \rightarrow 2$

and the  $1 \rightarrow 4$  transitions in a 20 nm GaAs quantum well with infinite barriers. What is the wavelength of the  $1 \rightarrow 2$  transition? ( $m_e^* = 0.067m_0$ .)

- (6.21) Linearly polarized light is incident on a quantum well sample at an angle  $\theta$  to the normal ( $z$ ) direction. The polarization direction lies within the plane of incidence. What is the maximum fraction of the power in the beam that can be absorbed by intersubband transitions if the refractive index of the crystal is 3.3?

- (6.22) Write down the energies of the first seven quantized energies of a cubic quantum dot with infinite barriers of dimension  $d$ . What is the degeneracy of each level?

- (6.23) Show that  $R(r) = \sin kr/r$  is a solution of the radial equation of a spherical dot for states with  $l = 0$ . By requiring that  $R(R_0) = 0$  in a dot with infinite barriers, show that the confinement energy of states with  $l = 0$  is equal to  $(\hbar^2/2m^*)(n\pi/R_0)^2$ , where  $n$  is an integer.

- (6.24) Compare the confinement energies of the ground states of cubic and spherical dots of the same volume and with infinite potential barriers in both cases.

- (6.25) Consider a two-dimensional harmonic oscillator with a Schrödinger equation given by:

$$\left[ -\frac{\hbar^2}{2m_e} \nabla_{2D}^2 + \frac{1}{2} m_e \omega_0^2 r^2 \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}),$$

with  $r^2 = x^2 + y^2$ . The form of the operator  $\nabla_{2D}^2$  is:

$$\nabla_{2D}^2 = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2}$$

in polar co-ordinates, and

$$\nabla_{2D}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

in Cartesian co-ordinates.

- (a) By working in Cartesian co-ordinates, show that the Schrödinger equation separates into two quantum harmonic oscillator equations for the  $x$  and  $y$  motion, and hence that:

$$E = (n+1)\hbar\omega_0,$$

where  $n = n_x + n_y$ ,  $n_x$  and  $n_y$  being the quantum numbers for the  $x$  and  $y$  axis harmonic oscillators. Hence write down the first five energy levels and their degeneracies.

- (b) By working in polar co-ordinates, show that the solutions take the form  $\psi(r, \phi) = R(r)e^{im\phi}$ ,

where  $m$  is an integer.

(c)\* The wave functions of a 1-D harmonic oscillator are given by:

$$\psi_n(x) = \left( \frac{1}{\sqrt{\pi} 2^n n!} \right)^{1/2} H_n(\xi) e^{-\xi^2/2},$$

where  $\xi = \sqrt{\alpha}x$ ,  $\alpha = m_e\omega_0/\hbar$ , and  $H_n(\xi)$  are the Hermite polynomials with  $H_0(\xi) = 1$ ,  $H_1(\xi) = 2\xi$ ,  $H_2(\xi) = 2 - 4\xi^2$ , etc. Similarly, the first six wave functions  $\psi_{n,m}(r, \phi)$  of the 2-D harmonic oscillator

in polar co-ordinates are of the form:

$$\begin{aligned}\psi_{0,0}(r, \phi) &\propto \exp(-\alpha r^2/2) \\ \psi_{1,\pm 1}(r, \phi) &\propto r \exp(-\alpha r^2/2) e^{\pm i\phi} \\ \psi_{2,0}(r, \phi) &\propto (\alpha r^2 - 1) \exp(-\alpha r^2/2) \\ \psi_{2,\pm 2}(r, \phi) &\propto r^2 \exp(-\alpha r^2/2) e^{\pm i2\phi}.\end{aligned}$$

By comparing the wave functions with the same energies in polar and Cartesian co-ordinates, verify that the allowed values of  $m$  for the first three levels are as shown in Fig. 6.19.